

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12404428
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Fukutomi; Shuhei et al.

Radiation-curable adhesive sheet

Abstract

A radiation curable pressure-sensitive adhesive sheet **1** of the present invention has a pressure-sensitive adhesive layer **10** that is cured by irradiation with radiation. The pressure-sensitive adhesive layer **10** contains a photopolymerization initiator **11a** and a crosslinking agent **11b**, and the curing by irradiation with radiation is curing by a reaction between the photopolymerization initiator **11a** and the crosslinking agent **11b**. In the radiation curable pressure-sensitive adhesive sheet **1** of the present invention, when the pressure-sensitive adhesive layer **10** is stored at 50° C. for 4 weeks, a rate of change in residual stress (N/cm.sup.2) expressed by the following equation is 70% or less.

$$\text{Residual stress change rate (\%)} = (\text{residual stress after storage at 50}^\circ \text{ C. for 4 weeks} - \text{initial residual stress}) / (\text{initial residual stress}) \times 100$$

Inventors:	Fukutomi; Shuhei (Ibaraki, JP), Katami; Hirofumi (Ibaraki, JP), Yamamoto; Shinya (Ibaraki, JP)
Applicant:	NITTO DENKO CORPORATION (Ibaraki, JP)
Family ID:	83458792
Assignee:	NITTO DENKO CORPORATION (Osaka, JP)
Appl. No.:	18/284135
Filed (or PCT Filed):	March 24, 2022
PCT No.:	PCT/JP2022/014047
PCT Pub. No.:	WO2022/210277
PCT Pub. Date:	October 06, 2022

Prior Publication Data

Document Identifier	Publication Date
----------------------------	-------------------------

Foreign Application Priority Data

JP 2021-058810 Mar. 30, 2021

Publication Classification

Int. Cl.: C09J7/38 (20180101); C09J11/06 (20060101); C09J133/14 (20060101)

U.S. Cl.:

CPC C09J7/385 (20180101); C09J11/06 (20130101); C09J133/14 (20130101);

Field of Classification Search

USPC: None

References Cited**FOREIGN PATENT DOCUMENTS**

Patent No.	Application Date	Country	CPC
106543907	12/2016	CN	N/A
106752992	12/2016	CN	N/A
109796880	12/2018	CN	N/A
2009-161603	12/2008	JP	N/A
2016-155981	12/2015	JP	N/A
2017-155213	12/2016	JP	N/A
2019-131679	12/2018	JP	N/A
2020-111628	12/2019	JP	N/A
2020-118918	12/2019	JP	N/A
2020-186312	12/2019	JP	N/A
2016/117045	12/2015	WO	N/A
2016/170875	12/2015	WO	N/A

OTHER PUBLICATIONS

JP2020118918A Machine Translation via EPO (Year: 2020). cited by examiner

International Search Report dated Jun. 7, 2022 in Application No. PCT/JP2022/014047. cited by applicant

Written Opinion of International Searching Authority dated Oct. 3, 2023 in Application No. PCT/JP2022/014047. cited by applicant

Notice of Reasons for Refusal dated Oct. 29, 2024 from the Japanese Patent Office in Application No. 2021-058810. cited by applicant

Extended European Search Report issued Feb. 24, 2025 in Application No. 22780491.1. cited by applicant

Communication issued Mar. 18, 2025 in Japanese Application No. 2021-058810. cited by applicant

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a National Stage of International Application No. PCT/JP2022/014047 filed Mar. 24, 2022, claiming priority based on Japanese Patent Application No. 2021-058810 filed Mar. 30, 2021.

TECHNICAL FIELD

(2) The present invention relates to a radiation curable pressure-sensitive adhesive sheet. In particular, the present invention relates to a radiation curable pressure-sensitive adhesive sheet that can be used for bonding a transparent optical element to another optical element.

BACKGROUND ART

(3) An image display device, such as a liquid crystal display device or an organic EL display device, is composed of an optical element laminate in which one of various types of transparent optical elements, such as a polarizing film, a retardation film, and a transparent cover element including a cover glass, is laminated. A pressure-sensitive adhesive sheet comprising a transparent pressure-sensitive adhesive layer is used for bonding these optical elements to each other. In other words, a pressure-sensitive adhesive sheet is disposed between two optical elements to be bonded together, and then the two optical elements are pressed against each other, so that they are bonded together through the pressure-sensitive adhesive sheet to form an optical element laminate. A pressure-sensitive adhesive sheet provided with the pressure-sensitive adhesive layer on one side of a substrate film is generally used in production processes of an optical product as a surface protective film to prevent scratches or adherence of stain on the optical element.

(4) For example, in an image display device equipped with an input device such as a touch panel, a transparent electroconductive printed layer, such as a patterned ITO (Indium Tin Oxide) layer, is formed on a surface of an optical element. Silver or copper lead wiring is further formed in the peripheral portion. In addition, a black concealing portion is generally printed in a frame shape in the peripheral edge portion of the transparent cover element. A pressure-sensitive adhesive sheet to which an optical element having such a printed layer and wiring is bonded is required to increase the fluidity of the pressure-sensitive adhesive layer to exhibit level difference absorbability such that no bubbles are left in the stepped space. In addition, when optical elements with printed stepped spaces are laminated together via a pressure-sensitive adhesive sheet, there is a problem that display unevenness is likely to occur in a peripheral edge portion of an image display panel due to stress strain or the like applied to the pressure-sensitive adhesive around the printed stepped spaces, and thus excellent stress relaxation properties are also required.

(5) On the other hand, a plastic film or the like used as an optical element contains a gas such as carbon dioxide, and gas may be generated under high-temperature conditions in the production process. At this point, when the pressure-sensitive adhesive layer is soft, the gas generation cannot be suppressed enough, and the pressure-sensitive adhesive layer is floated, thus causing a problem that bubbles are likely to be formed. Therefore, in order to suppress the gas generation from the plastic film, the pressure-sensitive adhesive sheet is required to have a high elastic modulus of the pressure-sensitive adhesive layer and be hard to enhance adhesive reliability.

(6) A pressure-sensitive adhesive sheet having a pressure-sensitive adhesive layer that is cured by irradiation with radiation (sometimes referred to herein as “radiation curable pressure-sensitive adhesive sheet”) is widely used as a pressure-sensitive adhesive sheet exhibiting all of above level difference absorbability, stress relaxation properties, and adhesive reliability (see, for example, Patent Literature 1). The radiation curable pressure-sensitive adhesive sheet has the advantage that,

before curing, the sheet can be made to have high fluidity and excellent level difference absorbability and stress relaxation properties to sufficiently follow the stepped space and reduce stress strain caused by the stepped space to suppress display unevenness, and then irradiation with radiation completes the curing process to improve adhesive reliability.

CITATION LIST

Patent Literature

(7) Patent Literature 1: International Publication No. WO 2016/170875

SUMMARY OF INVENTION

Technical Problem

(8) Although the radiation curable pressure-sensitive adhesive sheet may be shipped as an pressure-sensitive adhesive sheet containing a photopolymerization initiator and a crosslinking agent and stored for a relatively long period of time, there has been a problem that curing gradually proceeds during storage due to environmental factors such as heat and light, resulting in deterioration of level difference absorbability and stress relaxation properties as well as generation of bubbles and display unevenness.

(9) The present invention has been made under the circumstances described above, and an object thereof is to provide a radiation curable pressure-sensitive adhesive sheet with excellent storage stability, which can suppress the progress of curing during storage and maintain excellent level difference absorbability and stress relaxation properties for a long time.

Solution to Problems

(10) As a result of intensive studies to achieve the object, the present inventors have found that a radiation curable pressure-sensitive adhesive sheet can suppress the progress of curing during storage, can maintain excellent level difference absorbability and stress relaxation properties for a long time, and has excellent storage stability by including a photopolymerization initiator exhibiting a pKa within a specific range as the photopolymerization initiator contained in the radiation curable pressure-sensitive adhesive sheet. The present invention has been completed based on these findings.

(11) In other words, the first aspect of the present invention provides a radiation curable pressure-sensitive adhesive sheet. The radiation curable pressure-sensitive adhesive sheet according to the first aspect of the present invention has a pressure-sensitive adhesive layer that is cured by irradiation with radiation, wherein: the pressure-sensitive adhesive layer contains a photopolymerization initiator and a crosslinking agent; and the curing by irradiation with radiation is curing by a reaction between the photopolymerization initiator and the crosslinking agent. In the radiation curable pressure-sensitive adhesive sheet according to the first aspect of the present invention, the pressure-sensitive adhesive layer is in a state of high fluidity, excellent level difference absorbability, and excellent stress relaxation properties before curing. Therefore, the pressure-sensitive adhesive layer can sufficiently follow a stepped space of a transparent electroconductive printed layer, such as a patterned ITO, formed on a surface of an optical element, silver or copper lead wiring, and a black concealing portion formed in a frame shape in a peripheral edge portion of a transparent cover element to be filled without bubbles remaining or gaps and can reduce stress strain caused by the stepped space to suppress display unevenness. The pressure-sensitive adhesive layer has the advantage that curing proceeds upon subsequent irradiation with radiation by a reaction between the photopolymerization initiator and the crosslinking agent contained in pressure-sensitive adhesive layer, thereby improving adhesive reliability.

(12) In the radiation curable pressure-sensitive adhesive sheet according to the first aspect of the present invention, when the pressure-sensitive adhesive layer is stored at 50° C. for 4 weeks, the rate of change in residual stress (N/cm.^{sup.2}) (hereinafter, sometimes referred to herein simply as “residual stress change rate”) expressed by the following equation is 70% or less.

Residual stress change rate (%)=(residual stress after storage at 50° C. for 4 weeks–initial residual stress)/(initial residual stress)×100

(13) When the radiation curable pressure-sensitive adhesive sheet is stored for a relatively long period of time, there is a problem that curing by a reaction of a photopolymerization initiator or a crosslinking agent contained in the pressure-sensitive adhesive layer gradually proceeds due to an environmental factor such as heat or light, resulting in deterioration of level difference absorbability and stress relaxation properties. The configuration in which the residual stress change rate is 70% or less is suitable in terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer during storage of the radiation curable pressure-sensitive adhesive sheet according to the first aspect of the present invention, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability.

(14) In the radiation curable pressure-sensitive adhesive sheet according to the first aspect of the present invention, in terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer during storage, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability, the residual stress change rate is preferably 60% or less and more preferably 55% or less, and may be 50% or less, 45% or less, 40% or less, 35% or less, or 30% or less.

Advantageous Effects of Invention

(15) A radiation curable pressure-sensitive adhesive sheet of the present invention can suppress the progress of curing during storage and maintain excellent level difference absorbability and stress relaxation properties for a long time, and has excellent storage stability.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a cross-sectional view illustrating one embodiment of a radiation curable pressure-sensitive adhesive sheet according to the present invention.

(2) FIGS. 2(a)-2(e) illustrate a process for implementing one embodiment of a method for producing a radiation curable pressure-sensitive adhesive sheet according to the present invention, in which FIG. 2(a) is a schematic view illustrating a pressure-sensitive adhesive layer forming step, FIG. 2(b) is a schematic view illustrating a pressure-sensitive adhesive layer curing step, FIG. 2(c) is a schematic view illustrating a solution application step, FIG. 2(d) is a schematic view illustrating a solution infiltration step, and FIG. 2(e) is a schematic view illustrating a drying step.

(3) FIG. 3 is a cross-sectional view illustrating one embodiment of the radiation curable pressure-sensitive adhesive sheet according to the present invention produced by the method in FIGS.

2(a)-2(e).

(4) FIG. 4 is a cross-sectional view of an optical element laminate presented as one example of the simplest embodiment using a radiation curable pressure-sensitive adhesive sheet according to the present invention.

(5) FIGS. 5(a)-5(d) are schematic views illustrating a process for implementing one embodiment of a method for producing an optical element laminate using the radiation curable pressure-sensitive adhesive sheet of the present invention illustrated in FIG. 3.

DESCRIPTION OF EMBODIMENTS

(6) The first aspect of the present invention provides a radiation curable pressure-sensitive adhesive sheet having a pressure-sensitive adhesive layer that is cured by irradiation with radiation. The pressure-sensitive adhesive layer contains a photopolymerization initiator and a crosslinking agent, and the curing by irradiation with radiation is curing by a reaction between the photopolymerization initiator and the crosslinking agent. When the pressure-sensitive adhesive layer is stored at 50° C. for 4 weeks, the rate of change in residual stress (N/cm.^{sup.2}) (hereinafter, sometimes referred to herein simply as “residual stress change rate”) expressed by the following

equation is 70% or less.

Residual stress change rate (%)=(residual stress after storage at 50° C. for 4 weeks–initial residual stress)/(initial residual stress)×100

(7) Hereinafter, in the present specification, the radiation curable pressure-sensitive adhesive sheet according to the first aspect of the present invention is sometimes referred to as “radiation curable pressure-sensitive adhesive sheet of the present invention” or simply as “pressure-sensitive adhesive sheet of the present invention”. The pressure-sensitive adhesive layer of the radiation curable pressure-sensitive adhesive sheet of the present invention is sometimes referred to as “pressure-sensitive adhesive layer of the present invention”, and the photopolymerization initiator and the crosslinking agent contained in the pressure-sensitive adhesive layer of the present invention are sometimes referred to as “photopolymerization initiator of the present invention” and “crosslinking agent of the present invention”, respectively.

(8) The form of the radiation curable pressure-sensitive adhesive sheet of the present invention is not particularly limited as long as the adhesive face is an adhesive face (pressure-sensitive adhesive layer surface) of the pressure-sensitive adhesive layer of the present invention. For example, the pressure-sensitive adhesive sheet may be a single-coated pressure-sensitive adhesive sheet having a pressure-sensitive adhesive surface on only one side thereof or a double-coated pressure-sensitive adhesive sheet having pressure-sensitive adhesive surfaces on both sides thereof. In addition, when the radiation curable pressure-sensitive adhesive sheet of the present invention is a double-coated pressure-sensitive adhesive sheet, the radiation curable pressure-sensitive adhesive sheet of the present invention may have a form in which both pressure-sensitive adhesive surfaces are provided by the pressure-sensitive adhesive layer of the present invention or may have a form in which one adhesive face is provided by the pressure-sensitive adhesive layer of the present invention while the other adhesive face is provided by a pressure-sensitive adhesive layer other than the pressure-sensitive adhesive layer of the present invention (other pressure-sensitive adhesive layer). From the viewpoint of laminating objects together, a double-coated pressure-sensitive adhesive sheet is preferable.

(9) The radiation curable pressure-sensitive adhesive sheet of the present invention may be a so-called “substrate-less type” of pressure-sensitive adhesive sheet having no substrate (substrate layer) or a type of a pressure-sensitive adhesive sheet having a substrate. Note that in the present specification, the “substrate-less type” of the pressure-sensitive adhesive sheet is sometimes referred to as a “substrate-less pressure-sensitive adhesive sheet”, and the type of the pressure-sensitive adhesive sheet having a substrate is sometimes referred to as a “substrate-attached pressure-sensitive adhesive sheet”. Examples of the above substrate-less pressure-sensitive adhesive sheet include a double-coated pressure-sensitive adhesive sheet composed only of the pressure-sensitive adhesive layer of the present invention and a double-coated pressure-sensitive adhesive sheet composed of the pressure-sensitive adhesive layer of the present invention and another pressure-sensitive adhesive layer (pressure-sensitive adhesive layer other than the pressure-sensitive adhesive layer of the present invention). Examples of the above substrate-attached pressure-sensitive adhesive sheet include a single-coated pressure-sensitive adhesive sheet having the pressure-sensitive adhesive layer of the present invention on one side of a substrate, a double-coated pressure-sensitive adhesive sheet having the pressure-sensitive adhesive layer of the present invention on both sides of a substrate, and a double-coated pressure-sensitive adhesive sheet having the pressure-sensitive adhesive layer of the present invention on one side of a substrate and another pressure-sensitive adhesive layer on the other side. Note that the above “substrate (substrate layer)” refers to a support, which is a portion to be laminated to an object together with a pressure-sensitive adhesive layer when the radiation curable pressure-sensitive adhesive sheet of the present invention is used (laminated) on the object. A separator (release liner) to be peeled off at the time of use (lamination) of the pressure-sensitive adhesive sheet is not included in the above substrate.

(10) Hereinafter, embodiments of the present invention will be described with reference to the

accompanying figures, but the present invention is not limited thereto. The embodiments are merely illustrative.

(11) FIG. 1 is a cross-sectional view illustrating one embodiment of a radiation curable pressure-sensitive adhesive sheet according to the present invention.

(12) A radiation curable pressure-sensitive adhesive sheet **1** according to one embodiment of the present invention in FIG. 1 is composed of: a pressure-sensitive adhesive layer **10**; a release sheet **S1** laminated onto one of opposite principal surfaces **10A** of the pressure-sensitive adhesive layer **10**; and a release sheet **S2** laminated onto the other principal surface **10B** of the pressure-sensitive adhesive layer **10**.

(13) In FIG. 1, the pressure-sensitive adhesive layer **10** comprises a base pressure-sensitive adhesive material and two opposite principal surfaces (first principal surface **10A** and second principal surface **10B**) and may be a single layer or have a laminated structure of two or more layers.

(14) The thickness of the pressure-sensitive adhesive layer **10** is not particularly limited but is typically 5 μm to 500 μm , preferably 5 μm to 400 μm , even more preferably 5 μm to 350 μm . If the thickness of the pressure-sensitive adhesive layer **10** is in this range, it is preferable in that the pressure-sensitive adhesive layer **10** can follow such that no bubbles are left in the stepped space, can reduce stress strain caused by the stepped space to suppress display unevenness, and has excellent level difference absorbability and stress relaxation properties.

(15) As measured according to JIS K7361, total light transmittance of the entire pressure-sensitive adhesive layer **10** is not particularly limited but is preferably 80% or more and preferably 90% or more. The higher total light transmittance of the pressure-sensitive adhesive layer **10** provides better results. Furthermore, a haze value thereof is preferably 1.5% or less, more preferably 1% or less, and 0.8% or less.

(16) The pressure-sensitive adhesive layer **10** contains a photopolymerization initiator **11a** and a crosslinking agent **11b**. The photopolymerization initiator **11a** and the crosslinking agent **11b** are preferably dissolved in the pressure-sensitive adhesive layer **10**. Herein, “dissolution” means, for example, that the photopolymerization initiator **11a** and the crosslinking agent **11b** dissolve to such an extent that the transparency of the pressure-sensitive adhesive layer **10** can be maintained, i.e., white turbidity caused by light scattering does not occur. Specifically, the photopolymerization initiator **11a** and the crosslinking agent **11b** are preferably contained in the pressure-sensitive adhesive layer **10** so that the haze value thereof is 1.5% or less, preferably 1% or less, and 0.8% or less. The present invention encompasses an embodiment in which either one of the photopolymerization initiator **11a** and the crosslinking agent **11b** is dissolved in the pressure-sensitive adhesive layer **10**.

(17) The photopolymerization initiator **11a** contained in the pressure-sensitive adhesive layer **10** has a pKa in the range of 5 to 12.7. When the photopolymerization initiator **11a** has a pKa in the range of 5 to 12.7, the pressure-sensitive adhesive layer **10** can suppress the progress of curing during storage, can maintain excellent level difference absorbability and stress relaxation properties for a long time, and has excellent storage stability.

(18) When the pressure-sensitive adhesive layer **10** is stored at 50° C. for 4 weeks, the rate of change in residual stress (N/cm.sup.2) (sometimes referred to herein simply as “residual stress change rate”) expressed by the following equation is 70% or less.

Residual stress change rate (%)=(residual stress after storage at 50° C. for 4 weeks–initial residual stress)/(initial residual stress)×100

(19) The configuration in which the rate of change in residual stress is 70% or less is preferable in terms of being able to maintain excellent level difference absorbability and stress relaxation properties when the radiation curable pressure-sensitive adhesive sheet **1** is used.

(20) Hereinafter, each configuration will be described in detail.

(21) <Pressure-Sensitive Adhesive Layer>

(22) The pressure-sensitive adhesive layer of the present invention is cured by irradiation with radiation, and the photopolymerization initiator of the present invention and the crosslinking agent of the present invention cause a polymerization reaction or a crosslinking reaction by irradiation with radiation to be cured. The pressure-sensitive adhesive layer is in a state of high fluidity, excellent level difference absorbability, and excellent stress relaxation properties before curing. Therefore, the pressure-sensitive adhesive layer of the present invention can sufficiently follow a stepped space of a transparent electroconductive printed layer, such as a patterned ITO, formed on a surface of an optical element, silver or copper lead wiring, and a black concealing portion formed in a frame shape in a peripheral edge portion of a transparent cover element to be filled without bubbles remaining or gaps and can reduce stress strain caused by the stepped space to suppress display unevenness. The pressure-sensitive adhesive layer of the present invention has the advantage that curing proceeds upon subsequent irradiation with radiation by a reaction between the photopolymerization initiator of the present invention and the crosslinking agent of the present invention contained in pressure-sensitive adhesive layer, thereby improving adhesive reliability.

(23) When the pressure-sensitive adhesive layer is stored at 50° C. for 4 weeks, the rate of change in residual stress (N/cm.sup.2) expressed by the following equation is 70% or less.

Residual stress change rate (%)=(residual stress after storage at 50° C. for 4 weeks–initial residual stress)/(initial residual stress)×100

(24) When the radiation curable pressure-sensitive adhesive sheet is stored for a relatively long period of time, there is a problem that curing by a reaction of a photopolymerization initiator or a crosslinking agent contained in the pressure-sensitive adhesive layer gradually proceeds due to an environmental factor such as heat or light, resulting in deterioration of level difference absorbability and stress relaxation properties. The configuration in which the residual stress change rate is 70% or less is suitable in terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer of the present invention during storage of the radiation curable pressure-sensitive adhesive sheet of the present invention, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability.

(25) In terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer of the present invention during storage, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability, the residual stress change rate is preferably 60% or less and more preferably 55% or less, and may be 50% or less, 45% or less, 40% or less, 35% or less, or 30% or less. It is most preferable that the lower limit of the residual stress change rate does not change, i.e., the residual stress change rate is 0%, but it can be said that the storage stability is excellent if the rate is about 1%.

(26) The pressure-sensitive adhesive layer of the present invention preferably has an initial residual stress of 13 N/cm.sup.2 or less at room temperature (e.g., 10 to 25° C., especially 25° C.). The configuration in which the pressure-sensitive adhesive layer of the present invention has an initial residual stress of 13 N/cm.sup.2 or less at room temperature is preferable in that the pressure-sensitive adhesive layer of the present invention has excellent level difference absorbability and stress relaxation properties. In terms of excellent level difference absorbability and stress relaxation properties of the pressure-sensitive adhesive layer of the present invention, the initial residual stress at room temperature is more preferably 12 N/cm.sup.2 or less, and may be 11 N/cm.sup.2 or less, 10 N/cm.sup.2 or less, 9 N/cm.sup.2 or less, 8 N/cm.sup.2 or less, 7 N/cm.sup.2 or less, 6 N/cm.sup.2 or less, 5 N/cm.sup.2 or less, or 4 N/cm.sup.2 or less.

(27) The pressure-sensitive adhesive layer of the present invention preferably has an initial residual stress of 13 N/cm.sup.2 or less at 40° C. The configuration in which the pressure-sensitive adhesive layer of the present invention has an initial residual stress of 13 N/cm.sup.2 or less at 40° C. is preferable in that the pressure-sensitive adhesive layer of the present invention has excellent level difference absorbability and stress relaxation properties. In terms of excellent level difference

absorbability and stress relaxation properties of the pressure-sensitive adhesive layer of the present invention, the initial residual stress at 40° C. is more preferably 12 N/cm.² or less, and may be 11 N/cm.² or less, 10 N/cm.² or less, 9 N/cm.² or less, 8 N/cm.² or less, 7 N/cm.² or less, 6 N/cm.² or less, 5 N/cm.² or less, 4 N/cm.² or less, or 3 N/cm.² or less.

(28) The pressure-sensitive adhesive layer of the present invention preferably has an initial residual stress of 15.5 N/cm.² or less at 0° C. The configuration in which the pressure-sensitive adhesive layer of the present invention has an initial residual stress of 15.5 N/cm.² or less at 0° C. is preferable in that the pressure-sensitive adhesive layer of the present invention has excellent level difference absorbability and stress relaxation properties. In terms of excellent level difference absorbability and stress relaxation properties of the pressure-sensitive adhesive layer of the present invention, the initial residual stress at 0° C. is more preferably 15 N/cm.² or less, 14 N/cm.² or less, 13 N/cm.² or less, 12 N/cm.² or less, and may be 11 N/cm.² or less, 10 N/cm.² or less, 9 N/cm.² or less, 8 N/cm.² or less, or 7 N/cm.² or less.

(29) The residual stress change rate and residual stress at room temperature, 40° C., or 0° C. in the pressure-sensitive adhesive layer of the present invention can be adjusted to desired ranges by adjusting the composition of monomer components of a base pressure-sensitive adhesive material contained in a pressure-sensitive adhesive composition for forming the pressure-sensitive adhesive layer described later, the type and blending amount of crosslinking agents (including first and second crosslinking agents described later), the type, blending amount, and pKa of polymerization initiators (including first and second photopolymerization initiators described later), and the type, blending amount, and pKa of other additives such as hindered amine light stabilizers.

(30) The base pressure-sensitive adhesive material contained in the pressure-sensitive adhesive composition for forming the pressure-sensitive adhesive layer of the present invention is not particularly limited as long as it is an adherent material usable in optical applications. For example, it is possible to use one or more appropriately selected from an acrylic pressure-sensitive adhesive, a rubber-based pressure-sensitive adhesive, a silicone-based pressure-sensitive adhesive, a polyester-based pressure-sensitive adhesive, a urethane-based pressure-sensitive adhesive, an epoxy-based pressure-sensitive adhesive, and a polyether-based pressure-sensitive adhesive. From the viewpoint of transparency, processability, durability, etc., it is preferable to use an acrylic pressure-sensitive adhesive. As the base pressure-sensitive adhesive material, the above pressure-sensitive adhesives can be used independently or in combination of two or more thereof. An acrylic polymer to be used as a base polymer of an acrylic pressure-sensitive adhesive is preferably, but not particularly limited to, a homopolymer or a copolymer of monomers containing a primary component consisting of (meth)acrylic acid alkyl ester. The expression “(meth)acrylic” is used herein to mean either one or both of “acrylic” and “methacrylic”, and the same applies to the other. In the present invention, the term “acrylic polymer” is used to mean that it may include the above (meth)acrylic acid alkyl ester and another monomer copolymerizable with it.

(31) When the base pressure-sensitive adhesive material contains an acrylic polymer as an acrylic pressure-sensitive adhesive, the acrylic polymer preferably contains a monomer unit derived from an acrylic acid alkyl ester having a linear or branched alkyl group and/or a methacrylic acid alkyl ester having a linear or branched alkyl group as the main monomer unit in the highest proportion by weight.

(32) Examples of the (meth)acrylic acid alkyl ester having a linear or branched alkyl group to constitute the monomer unit of the acrylic polymer, i.e., the (meth)acrylic acid alkyl ester having a linear or branched alkyl group contained in a monomer component to form the acrylic polymer, include a (meth)acrylic acid alkyl ester having a linear or branched alkyl group with 1 to 20 carbon atoms, such as methyl (meth)acrylate, ethyl (meth)acrylate, propyl (meth)acrylate, isopropyl (meth)acrylate, n-butyl (meth)acrylate, s-butyl (meth)acrylate, isobutyl (meth)acrylate, t-butyl (meth)acrylate, pentyl (meth)acrylate, isopentyl (meth)acrylate, hexyl (meth)acrylate, heptyl

(meth)acrylate, octyl (meth)acrylate, 2-ethylhexyl (meth)acrylate, isooctyl (meth)acrylate, nonyl (meth)acrylate, isononyl (meth)acrylate, decyl (meth)acrylate, isodecyl (meth)acrylate, undecyl (meth)acrylate, dodecyl (meth)acrylate, tridecyl (meth)acrylate, tetradecyl (meth)acrylate, pentadecyl (meth)acrylate, hexadecyl (meth)acrylate, heptadecyl (meth)acrylate, octadecyl (meth)acrylate, isostearyl (meth)acrylate, nonadecyl (meth)acrylate, and eicosyl (meth)acrylate. As the (meth)acrylic acid alkyl ester for the acrylic polymer, one (meth)acrylic acid alkyl ester may be used, or two or more (meth)acrylic acid alkyl esters may be used. In the present embodiment, at least one selected from the group consisting of n-butyl acrylate, 2-ethylhexyl acrylate, and isostearyl acrylate is preferably used as the (meth)acrylic acid alkyl ester for the acrylic polymer.

(33) In the acrylic polymer, a proportion of the monomer unit derived from the (meth)acrylic acid alkyl ester having a linear or branched alkyl group is preferably 50% by weight or more, more preferably 60% by weight or more, more preferably 70% by weight or more, more preferably 80% by weight or more, and more preferably 90% by weight or more. In other words, the proportion of the (meth)acrylic acid alkyl ester in a monomer composition of a raw material to form the acrylic polymer is preferably 50% by weight or more, more preferably 60% by weight or more, more preferably 70% by weight or more, more preferably 80% by weight or more, and more preferably 90% by weight or more.

(34) The acrylic polymer contained in the base pressure-sensitive adhesive material may contain a monomer unit derived from an alicyclic monomer. Examples of the alicyclic monomer to constitute the monomer unit of the acrylic polymer, i.e., the alicyclic monomer contained in the monomer component to form the acrylic polymer, include (meth)acrylic acid cycloalkyl ester, (meth)acrylic acid ester having a bicyclic hydrocarbon ring, and (meth)acrylic acid ester having a tricyclic or more hydrocarbon ring. Examples of the (meth)acrylic acid cycloalkyl ester include cyclopentyl (meth)acrylate, cyclohexyl (meth)acrylate, cycloheptyl (meth)acrylate, and cyclooctyl (meth)acrylate. Examples of the (meth)acrylic acid ester having bicyclic hydrocarbon ring include bornyl (meth)acrylate and isobornyl (meth)acrylate. Examples of the (meth)acrylic acid ester having a tricyclic or more hydrocarbon ring include dicyclopentanyl (meth)acrylate, dicyclopentanyloxyethyl (meth)acrylate, tricyclopentanyl (meth)acrylate, 1-adamantyl (meth)acrylate, 2-methyl-2-adamantyl (meth)acrylate, and 2-ethyl-2-adamantyl (meth)acrylate. As the alicyclic monomer for the acrylic polymer, one alicyclic monomer may be used, or two or more alicyclic monomers may be used. In the present embodiment, at least one selected from the group consisting of cyclohexyl acrylate, cyclohexyl methacrylate, isobornyl acrylate, and isobornyl methacrylate is preferably used as the alicyclic monomer for the acrylic polymer.

(35) In the acrylic polymer, a proportion of the monomer unit derived from the alicyclic monomer is preferably 5 to 60% by weight, more preferably 10 to 50% by weight, more preferably 12 to 40% by weight, from the viewpoint of realizing appropriate flexibility in the base pressure-sensitive adhesive material formed containing the acrylic polymer.

(36) The acrylic polymer contained in the base pressure-sensitive adhesive material may contain a monomer unit derived from a hydroxy group-containing monomer. The hydroxy group-containing monomer is a monomer having at least one hydroxy group in its monomer unit. When the acrylic polymer in the base pressure-sensitive adhesive material contains a hydroxy group-containing monomer unit, the base pressure-sensitive adhesive material readily offers adhesiveness and appropriate cohesive force. The hydroxy group may also serve as a reactive site to a crosslinking agent described later.

(37) Examples of the hydroxy group-containing monomer to constitute a monomer unit of the acrylic polymer, i.e., the hydroxy group-containing monomer contained in the monomer component to form the acrylic polymer, include hydroxy group-containing (meth)acrylic acid ester, vinyl alcohol, and allyl alcohol. Examples of the hydroxy group-containing (meth)acrylic acid ester include 2-hydroxyethyl (meth)acrylate, 2-hydroxypropyl (meth)acrylate, 3-hydroxypropyl (meth)acrylate, 4-hydroxybutyl (meth)acrylate, 6-hydroxyhexyl (meth)acrylate, hydroxyoctyl

(meth)acrylate, hydroxydecyl (meth)acrylate, hydroxylauryl (meth)acrylate, and (4-hydroxymethylcyclohexyl)methyl (meth)acrylate. As the hydroxy group-containing monomer for the acrylic polymer, one hydroxy group-containing monomer may be used, or two or more hydroxy group-containing monomers may be used. In the present embodiment, at least one selected from the group consisting of 2-hydroxyethyl acrylate, 2-hydroxyethyl methacrylate, 2-hydroxypropyl acrylate, 2-hydroxypropyl methacrylate, 4-hydroxybutyl acrylate, and 4-hydroxybutyl methacrylate is preferably used as the hydroxy group-containing monomer for the acrylic polymer.

(38) In the acrylic polymer, a proportion of the monomer unit derived from the hydroxy group-containing monomer is preferably 1% by weight or more, more preferably 2% by weight or more, more preferably 3% by weight or more, more preferably 7% by weight or more, more preferably 10% by weight or more, and more preferably 15% by weight or more. In the acrylic polymer, the proportion of the monomer unit derived from the hydroxy group-containing monomer is preferably 35% by weight or less and more preferably 30% by weight or less. These configurations regarding the proportion of the hydroxy group-containing monomer are preferable to realize adhesiveness and appropriate cohesive force in the base pressure-sensitive adhesive material formed containing the acrylic polymer.

(39) The acrylic polymer contained in the base pressure-sensitive adhesive material may contain a monomer unit derived from a nitrogen atom-containing monomer. The nitrogen atom-containing monomer is a monomer having at least one nitrogen atom in its monomer unit. When the acrylic polymer in the base pressure-sensitive adhesive material contains a nitrogen atom-containing monomer unit, the base pressure-sensitive adhesive material readily offers hardness and satisfactory adhesive reliability.

(40) Examples of the nitrogen atom-containing monomer to constitute a monomer unit of the acrylic polymer, i.e., the nitrogen atom-containing monomer contained in the monomer component to form the acrylic polymer, include cyclic N-vinylamides and (meth)acrylamides. Examples of the cyclic N-vinylamides which are nitrogen atom-containing monomers include N-vinyl-2-pyrrolidone, N-vinyl-2-piperidone, N-vinyl-3-morpholinone, N-vinyl-2-caprolactam, N-vinyl-1,3-oxazin-2-one, and N-vinyl-3,5-morpholinedione. Examples of the (meth)acrylamides which are nitrogen atom-containing monomers include (meth)acrylamide, N-ethyl (meth)acrylamide, N-isopropyl (meth)acrylamide, N-n-butyl (meth)acrylamide, N-octyl (meth)acrylamide, N,N-dimethyl (meth)acrylamide, N,N-diethyl (meth)acrylamide, N,N-dipropyl (meth)acrylamide, and N,N-diisopropyl (meth)acrylamide. As the nitrogen atom-containing monomer for the acrylic polymer, one nitrogen atom-containing monomer may be used, or two or more nitrogen atom-containing monomers may be used. In the present embodiment, N-vinyl-2-pyrrolidone is preferably used as the nitrogen atom-containing monomer for the acrylic polymer.

(41) In the acrylic polymer, a proportion of the monomer unit derived from the nitrogen atom-containing monomer is preferably 1% by weight or more, more preferably 3% by weight or more, and more preferably 5% by weight or more, from the viewpoint of realizing appropriate hardness, adhesiveness, and transparency in the base pressure-sensitive adhesive material formed containing the acrylic polymer. In the acrylic polymer, the proportion of the monomer unit derived from the nitrogen atom-containing monomer is preferably 30% by weight or less and more preferably 25% by weight or less, from the viewpoint of realizing sufficient transparency in the base pressure-sensitive adhesive material formed by containing the acrylic polymer and the viewpoint of suppressing the base pressure-sensitive adhesive material from becoming excessively hard to realize satisfactory adhesive reliability.

(42) The acrylic polymer contained in the base pressure-sensitive adhesive material may contain a monomer unit derived from a carboxy group-containing monomer. The carboxy group-containing monomer is a monomer having at least one carboxy group in its monomer unit. When the acrylic polymer in the base pressure-sensitive adhesive material contains a carboxy group-containing monomer unit, the base pressure-sensitive adhesive material may offer satisfactory adhesive

reliability. The carboxy group may also serve as a reactive site to a crosslinking agent described later.

(43) Examples of the carboxy group-containing monomer to constitute a monomer unit of the acrylic polymer, i.e., the carboxy group-containing monomer contained in the monomer component to form the acrylic polymer, include (meth)acrylic acid, itaconic acid, maleic acid, fumaric acid, crotonic acid, and isocrotonic acid. As the carboxy group-containing monomer for the acrylic polymer, one carboxy group-containing monomer may be used, or two or more carboxy group-containing monomers may be used. In the present embodiment, acrylic acid is preferably used as the carboxy group-containing monomer for the acrylic polymer.

(44) In the acrylic polymer, a proportion of the monomer unit derived from the carboxy group-containing monomer is preferably 0.1% by weight or more and more preferably 0.5% by weight or more, from the viewpoint of obtaining a contribution of interaction between a polar group and a carboxy group when the polar group is present on an object surface in the base pressure-sensitive adhesive material formed by containing the acrylic polymer to ensure satisfactory adhesive reliability. In the acrylic polymer, the proportion of the monomer unit derived from the carboxy group-containing monomer is also preferably 20% by weight or less and more preferably 15% by weight or less, from the viewpoint of suppressing the base pressure-sensitive adhesive material formed by containing the acrylic polymer from becoming excessively hard to realize satisfactory adhesive reliability.

(45) The acrylic polymer contained in the base pressure-sensitive adhesive material may have a crosslinked structure derived from a crosslinking agent. Having a crosslinked structure, the base pressure-sensitive adhesive material has an increased viscosity and thus making it easier to improve shape stability. Examples of the crosslinking agent include polyfunctional (meth)acrylate which is a copolymerizable crosslinking agent and a thermosetting crosslinking agent. The acrylic polymer may have a crosslinked structure derived only from the polyfunctional (meth)acrylate, a crosslinked structure derived only from the thermosetting crosslinking agent, and a crosslinked structure derived from both the polyfunctional (meth)acrylate and the thermosetting crosslinking agent.

(46) Note that the crosslinking agent has already formed a crosslinked structure by the progress of a crosslinking reaction in the acrylic polymer forming the pressure-sensitive adhesive layer of the present invention. On the other hand, the crosslinking agent of the present invention contained in the pressure-sensitive adhesive layer of the present invention is an unreacted crosslinking agent before forming a crosslinked structure. Therefore, the crosslinking agent forming a crosslinked structure in the acrylic polymer forming the pressure-sensitive adhesive layer of the present invention does not correspond to the crosslinking agent of the present invention contained in the pressure-sensitive adhesive layer of the present invention. In the present specification, the crosslinking agent forming a crosslinked structure in the acrylic polymer is sometimes referred to as a “first crosslinking agent”, and the crosslinking agent of the present invention is sometimes referred to as a “second crosslinking agent”.

(47) Examples of the polyfunctional (meth)acrylate as the first crosslinking agent include 1,6-hexanediol di(meth)acrylate, butanediol di(meth)acrylate, (poly)ethylene glycol di(meth)acrylate, (poly)propylene glycol di(meth)acrylate, neopentyl glycol di(meth)acrylate, pentaerythritol di(meth)acrylate, pentaerythritol tri(meth)acrylate, dipentaerythritol hexa(meth)acrylate, trimethylolpropane tri(meth)acrylate, tetramethylolmethane tri(meth)acrylate, allyl(meth)acrylate, and vinyl(meth)acrylate. As the polyfunctional (meth)acrylate for the acrylic polymer, one polyfunctional (meth)acrylate may be used, or two or more polyfunctional (meth)acrylates may be used. In the present embodiment, at least one selected from the group consisting of 1,6-hexanediol diacrylate, dipentaerythritol hexaacrylate, and trimethylolpropane triacrylate is preferably used as the polyfunctional (meth)acrylate for the acrylic polymer.

(48) In the acrylic polymer, a proportion of the monomer unit derived from the polyfunctional

(meth)acrylate as the first crosslinking agent is preferably 0.001% by weight or more and more preferably 0.005% by weight or more, and may be 0.01% by weight or more, 0.015% by weight or more, 0.02% by weight or more, 0.025% by weight or more, 0.03% by weight or more, 0.035% by weight or more, 0.04% by weight or more, 0.045% by weight or more, 0.05% by weight or more, 0.06% by weight or more, 0.07% by weight or more, 0.08% by weight or more, 0.09% by weight or more, or 0.1% by weight or more. In the acrylic polymer, the proportion of the monomer unit derived from the polyfunctional (meth)acrylate as the first crosslinking agent is preferably 1% by weight or less and more preferably 0.9% by weight or less, and may be 0.8% by weight or less, 0.7% by weight or less, 0.6% by weight or less, or 0.5% by weight or less. These configurations regarding the proportion of the polyfunctional (meth)acrylate as the first crosslinking agent are preferable to realize appropriate hardness, adhesiveness, and shape stability in the base pressure-sensitive adhesive material formed by containing the acrylic polymer.

(49) Examples of the thermosetting crosslinking agent as the first crosslinking agent include an isocyanate-based crosslinking agent, an epoxy-based crosslinking agent, a metal chelate-based crosslinking agent, a melamine-based crosslinking agent, a peroxide-based crosslinking agent, an urea-based crosslinking agent, a metal alkoxide-based crosslinking agent, a metal salt-based crosslinking agent, a carbodiimide-based crosslinking agent, an oxazoline-based crosslinking agent, an aziridine-based crosslinking agent, and an amine-based crosslinking agent. The base pressure-sensitive adhesive material may contain one of the thermosetting crosslinking agents or two or more of the thermosetting crosslinking agents. Preferably, at least one selected from the group consisting of an isocyanate-based crosslinking agent and an epoxy-based crosslinking agent is used.

(50) Examples of the isocyanate-based crosslinking agent include lower aliphatic polyisocyanates, alicyclic polyisocyanates, and aromatic polyisocyanates. Examples of the lower aliphatic polyisocyanates include 1,2-ethylene diisocyanate, 1,4-butylene diisocyanate, and 1,6-hexamethylene diisocyanate. Examples of the alicyclic polyisocyanates include cyclopentylene diisocyanate, cyclohexylene diisocyanate, isophorone diisocyanate, hydrogenated tolylene diisocyanate, and hydrogenated xylene diisocyanate. Examples of the aromatic polyisocyanates include 2,4-tolylene diisocyanate, 2,6-tolylene diisocyanate, 4,4'-diphenylmethane diisocyanate, and xylylene diisocyanate. Examples of the isocyanate-based crosslinking agent include commercial products such as trimethylolpropane/tolylene diisocyanate adduct (trade name "CORONATE L", manufactured by Tosoh Corporation), trimethylolpropane/hexamethylene diisocyanate adduct (trade name "CORONATE HL", manufactured by Tosoh Corporation), and trimethylolpropane/xylylene diisocyanate adduct (trade name "TAKENATE D-110N", manufactured by Mitsui Chemicals Inc.).

(51) Examples of the epoxy-based crosslinking agent (polyfunctional epoxy compound) include N,N,N',N'-tetraglycidyl-m-xylenediamine, diglycidylaniline, 1,3-bis(N,N-diglycidylaminomethyl)cyclohexane, 1,6-hexanediol diglycidyl ether, neopentyl glycol diglycidyl ether, ethylene glycol diglycidyl ether, propylene glycol diglycidyl ether, polyethylene glycol diglycidyl ether, polypropylene glycol diglycidyl ether, sorbitol polyglycidyl ether, glycerol polyglycidyl ether, pentaerythritol polyglycidyl ether, polyglycerol polyglycidyl ether, sorbitan polyglycidyl ether, trimethylolpropane polyglycidyl ether, adipic acid diglycidyl ester, o-phthalic acid diglycidyl ester, triglycidyl-tris(2-hydroxyethyl)isocyanurate, resorcin diglycidyl ether, and bisphenol-S-diglycidyl ether. Examples of the epoxy-based crosslinking agent include an epoxy-based resin having two or more epoxy groups. In addition, examples of the epoxy-based crosslinking agent include commercial products such as trade name "TETRAD C" (manufactured by Mitsubishi Gas Chemical Company, Inc.).

(52) When the base pressure-sensitive adhesive material contains the above thermosetting crosslinking agent as the first crosslinking agent for crosslinking between the acrylic polymers, a content of the thermosetting crosslinking agent in the base pressure-sensitive adhesive material is

preferably 0.001 parts by weight or more and more preferably 0.01 parts by weight or more relative to 100 parts by weight of the acrylic polymer in the base pressure-sensitive adhesive material, from the viewpoint of making it easier to improve the shape stability of the base pressure-sensitive adhesive material, and thus realizing sufficient adhesive reliability to the object. The content of the thermosetting crosslinking agent in the base pressure-sensitive adhesive material is also preferably 10 parts by weight or less and more preferably 5 parts by weight or less relative to 100 parts by weight of the acrylic polymer in the base pressure-sensitive adhesive material, from the viewpoint of imparting appropriate flexibility to the base pressure-sensitive adhesive material to realize satisfactory adhesive strength.

(53) When the base pressure-sensitive adhesive material contains the above acrylic polymer as a pressure-sensitive adhesive, a content ratio of the acrylic polymer in the base pressure-sensitive adhesive material is, for example, 85 to 100% by weight.

(54) The base pressure-sensitive adhesive material may also contain a polymerization initiator in addition to the monomer to form the acrylic polymer and the first crosslinking agent. Examples of the polymerization initiator include a photopolymerization initiator and a thermopolymerization initiator. The base pressure-sensitive adhesive material may contain one polymerization initiator or two or more polymerization initiators.

(55) Note that the polymerization initiator functions as a catalyst for promoting the polymerization and crosslinking reaction of the above monomer component and the crosslinking agent to form an acrylic polymer. Therefore, it is considered that the acrylic polymer is deactivated and decomposed when the acrylic polymer is formed and does not remain in the pressure-sensitive adhesive layer of the present invention or remains in a trace amount even if it remains. On the other hand, the photopolymerization initiator of the present invention is contained in the pressure-sensitive adhesive layer of the present invention together with the acrylic polymer and promotes the crosslinking reaction of the crosslinking agent of the present invention. Therefore, the photopolymerization initiator for forming the acrylic polymer does not correspond to the photopolymerization initiator of the present invention, and a photopolymerization initiator whose pKa does not correspond to 5 to 12.7 can also be used without limitation. In the present specification, the photopolymerization initiator for forming the acrylic polymer is sometimes referred to as a “first photopolymerization initiator”, and the photopolymerization initiator of the present invention is sometimes referred to as a “second photopolymerization initiator”.

(56) Examples of the first photopolymerization initiator include a benzoin ether-based photopolymerization initiator, an acetophenone-based photopolymerization initiator, an aminoacetophenone-based photopolymerization initiator, an α -ketol-based photopolymerization initiator, an aromatic sulfonyl chloride-based photopolymerization initiator, a photoactive oxime-based photopolymerization initiator, a benzoin-based photopolymerization initiator, a benzyl-based photopolymerization initiator, a benzophenone-based photopolymerization initiator, a ketal-based photopolymerization initiator, and a thioxanthone-based photopolymerization initiator. Examples of the benzoin ether-based photopolymerization initiator include benzoin methyl ether, benzoin ethyl ether, benzoin propyl ether, benzoin isopropyl ether, benzoin isobutyl ether, and 2,2-dimethoxy-1,2-diphenylethan-1-one. Examples of the acetophenone-based photopolymerization initiator include 2,2-diethoxyacetophenone, 2,2-dimethoxy-2-phenylacetophenone, 1-hydroxycyclohexyl phenyl ketone, 4-phenoxy dichloroacetophenone, and 4-(t-butyl)dichloroacetophenone. Examples of the aminoacetophenone-based photopolymerization initiator include 2-methyl-1-[4-(methylthio)phenyl]-2-morpholinopropan-1-one, 2-methyl-1-phenyl-2-morpholinopropan-1-one, 2-methyl-1-[4-(hexyl)phenyl]-2-morpholinopropan-1-one, 2-ethyl-2-dimethylamino-1-(4-morpholinophenyl)-butanone-1, 2-benzyl-2-dimethylamino-1-(4-morpholinophenyl)-butanone-1, and 2-(dimethylamino)-2-[(4-methylphenyl)methyl]-1-[4-(4-morpholinyl)phenyl]-1-butanone. Examples of the α -ketol-based photopolymerization initiator include 2-methyl-2-hydroxy propiophenone, 1-hydroxycyclohexyl phenyl ketone, 2-hydroxy-1-(4-(4-(2-hydroxy-2-

methylpropionyl)benzyl)phenyl)-2-methylpropan-1-one and 1-[4-(2-hydroxyethyl)phenyl]-2-methylpropan-1-one. Examples of the aromatic sulfonyl chloride-based photopolymerization initiator include 2-naphthalenesulfonyl chloride. Examples of the photoactive oxime-based photopolymerization initiator include 1-phenyl-1,1-propanedione-2-(o-ethoxycarbonyl)-oxime. Examples of the benzoin-based photopolymerization initiator include benzoin. Examples of the benzyl-based photopolymerization initiator include benzyl. Examples of the benzophenone-based photopolymerization initiator include benzophenone, benzoylbenzoic acid, 3,3'-dimethyl-4-methoxy benzophenone, and polyvinyl benzophenone. Examples of the ketal-based photopolymerization initiator include benzyl dimethyl ketal. Examples of the thioxanthone-based photopolymerization initiator include thioxanthone, 2-chloro thioxanthone, 2-methyl thioxanthone, 2,4-dimethyl thioxanthone, isopropyl thioxanthone, 2,4-diisopropyl thioxanthone, and dodecyl thioxanthone.

(57) An amount of the first photopolymerization initiator to be used is not particularly limited, but for example, it is preferably 0.001 to 1 part by weight and more preferably 0.01 to 0.50 parts by weight relative to 100 parts by weight of a total monomer unit of the acrylic polymer (total monomer component constituting the acrylic polymer).

(58) Examples of the thermopolymerization initiator include an azo-based polymerization initiator, a peroxide-based polymerization initiator, and a redox-based polymerization initiator. Examples of the azo-based polymerization initiator include 2,2'-azobisisobutyronitrile (AIBN), 2,2'-azobis-2-methylbutyronitrile (AMBN), 2,2'-azobis(2-methylpropionate)dimethyl, 4,4'-azobis-4-cyanovaleric acid, 2,2'-azobis(4-methoxy-2,4-dimethylvaleronitrile), 2,2'-azobis(2,4-dimethylvaleronitrile), 1,1'-azobis(cyclohexane-1-carbonitrile), and 2,2'-azobis(2,4,4-trimethylpentane). Examples of the peroxide-based polymerization initiator include benzoyl peroxide, t-butyl hydroperoxide, di-t-butyl peroxide, t-butyl peroxybenzoate, dicumyl peroxide, 1,1-bis(t-butyl peroxy)-3,3,5-trimethylcyclohexane, and 1,1-bis(t-butyl peroxy)cyclododecane.

(59) An amount of the thermopolymerization initiator to be used is not particularly limited, but for example, it is preferably 0.05 to 0.5 parts by weight and more preferably 0.1 to 0.3 parts by weight relative to 100 parts by weight of a total monomer unit of the acrylic polymer (total content of a monomer component constituting the acrylic polymer).

(60) The pressure-sensitive adhesive composition may further contain additives, in addition to the base pressure-sensitive adhesive material, such as an ultraviolet absorbing agent, a rust inhibitor, an antistatic agent, a crosslinking accelerator, a silane coupling agent, a tackifier resin, an anti-aging agent, a filler, a coloring agent including pigment and dye, an antioxidant, a chain transfer agent, a plasticizer, a softener, and a surfactant, if necessary. Examples of the tackifier resin include a rosin derivative, polyterpene resin, petroleum resin, and oil-soluble phenol.

(61) As the crosslinking agent of the present invention (second crosslinking agent) contained in the pressure-sensitive adhesive layer of the present invention, those illustrated as the above first crosslinking agent can be used without limitation, and a polyfunctional(meth)acrylate, which is a crosslinking agent having polymerization properties by irradiation with radiation (radiation curability) is preferable. As the second crosslinking agent, one of the crosslinking agent may be used, or two or more of the crosslinking agents may be used. In addition, the first crosslinking agent and the second crosslinking agent may be the same crosslinking agent or a combination of different crosslinking agents. In the present embodiment, trimethylolpropane tri(meth)acrylate, dipentaerythritol hexa(meth)acrylate, or 1,6-hexanediol di(meth)acrylate is preferably used as the crosslinking agent of the present invention (second crosslinking agent).

(62) The content of the crosslinking agent of the present invention (second crosslinking agent) contained in the pressure-sensitive adhesive layer of the present invention can be appropriately set as long as it can be cured by irradiation with radiation to provide excellent adhesive reliability, and is not particularly limited, but can be selected from the range of, for example, in terms of the upper limit, 20% by weight or less (e.g., 0.001 to 20% by weight, 0.001 to 18% by weight, 0.001 to 16%

by weight, 0.001 to 14% by weight, 0.001 to 12% by weight, 0.001 to 10% by weight, 0.001 to 8% by weight, and 0.001 to 6% by weight); or in terms of the lower limit, 0.001% or more (e.g., 0.001 to 20% by weight, 0.005 to 20% by weight, 0.01 to 20% by weight, 0.05 to 20% by weight, 0.1 to 20% by weight, 0.5 to 20% by weight, 1 to 20% by weight, 1.1 to 20% by weight, 1.2 to 20% by weight, and 1.3 to 20% by weight) relative to the pressure-sensitive adhesive layer of the present invention (100% by weight).

(63) The photopolymerization initiator of the present invention (second photopolymerization initiator) contained in the pressure-sensitive adhesive layer of the present invention is not particularly limited, but for example, a photopolymerization initiator having pKa of 5 to 12.7 can be used. By using the photopolymerization initiator having a pKa of 5 to 12.7, it is possible to suppress the progress of curing of the pressure-sensitive adhesive layer of the present invention during storage of the radiation curable pressure-sensitive adhesive sheet of the present invention, maintain excellent level difference absorbability and stress relaxation properties for a long time, and improve excellent storage stability. Of course, the effects of the present application can be exhibited even if the pKa is outside the above range.

(64) The pKa (acid dissociation constant) of the photopolymerization initiator is an index for quantitatively indicating the strength of an acid, and the smaller the pKa, the stronger the acidity. It is considered that when the pressure-sensitive adhesive has a high acid value, the second crosslinking agent (especially, polyfunctional (meth)acrylate) undergoes a Michael addition reaction, resulting in a gradual crosslinking reaction. When the pKa of the photopolymerization initiator of the present invention (second photopolymerization initiator) is in the range of 5 to 12.7, it is presumed that the acid value in the pressure-sensitive adhesive layer is adjusted to suppress the crosslinking reaction over time. Note that this mechanism is presumptive and should not be construed as limiting the scope of the present invention.

(65) In the radiation curable pressure-sensitive adhesive sheet of the present invention, the pKa of the photopolymerization initiator of the present invention is preferably 5.1 or more and more preferably 5.2 or more, and may be 5.3 or more or 5.4 or more in terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer during storage, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability. In terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer during storage, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability, the pKa of the photopolymerization initiator of the present invention is also preferably 12.5 or less and more preferably 12 or less, and may be 11.5 or less, 11 or less, 10.5 or less, 10 or less, 9.5 or less, 9 or less, 8.5 or less, or 8.0 or less.

(66) The pKa of the photopolymerization initiator can be measured by a known method such as neutralization titration, absorptiometry, or capillary electrophoresis, but may not be applicable in some cases depending on the physical properties of the photopolymerization initiator. Measurement errors may also occur due to measurement conditions such as temperature and concentration. In order to solve such a problem, it is preferable to use a calculated pKa determined from the chemical structural formula of the photopolymerization initiator. The calculated pKa can be easily and accurately determined regardless of the physical properties as long as the chemical structural formula is given, and measurement errors due to measurement conditions and the like do not occur, making it easy to compare photopolymerization initiators. Therefore, it is preferable to use the calculated pKa determined based on the chemical structural formula as the pKa of the photopolymerization initiator of the present invention.

(67) The calculated pKa is not particularly limited, and for example, a chemical structural formula is drawn using ChemDraw 19.0, and the chemical structural formula can be obtained using a computational module. As the computational module, for examples, the Chemoinformatics platform MOSES from Molecular Networks can be used. MOSES is a computational module

developed, maintained, and owned by Molecular Networks GmbH (Erlangen, Germany).

(68) As the photopolymerization initiator of the present invention (second photopolymerization initiator) contained in the pressure-sensitive adhesive layer of the present invention, among the photopolymerization initiator illustrated as the above first photopolymerization initiator, a photopolymerization initiator having a pKa of 5 to 12.7 can be used without limitation. Specifically, a chemical structural formula of the photopolymerization initiator is drawn using ChemDraw 19.0, and the chemical structural formula with a calculated pKa of 5 to 12.7 obtained using a computational module can be used. When the first photopolymerization initiator has a pKa of 5 to 12.7, the first photopolymerization initiator and the second photopolymerization initiator may be the same or a combination of different photopolymerization initiators.

(69) Specifically, the photopolymerization initiator of the present invention (second photopolymerization initiator) is preferably a photopolymerization initiator having an amino group and an aromatic ring in the molecule thereof and more preferably a photopolymerization initiator having two or more amino groups and an aromatic ring in the molecule thereof. The photopolymerization initiator of the present invention is preferably an aminoacetophenone-based photopolymerization initiator. The photopolymerization initiator having the chemical structure of the above configuration is considered to have a pKa that is easily controlled in the range of 5 to 12.7 due to the amino group and the aromatic ring in the chemical structure. Note that a photopolymerization initiator having a chemical structure with the above composition but not having a pKa in the range of 5 to 12.7 does not correspond to the photopolymerization initiator of the present invention.

(70) More specifically, the photopolymerization initiator of the present invention (second photopolymerization initiator) is preferably a photopolymerization initiator having a chemical structure of the following formula (1).

(71) ##STR00001##

(72) In the above formula (1), Ar represents a phenyl group substituted with —SR^{sup.5} or —N(R^{sup.6})(R^{sup.7}), and R^{sup.5} represents a hydrogen atom or an alkyl group.

(73) R^{sup.1} and R^{sup.2} each independently represent an alkyl group having 1 to 8 carbon atoms. R^{sup.1} and R^{sup.2} may bond with each other to form an alkylene group having 2 to 9 carbon atoms.

(74) The alkyl group represented by R^{sup.1} and R^{sup.2} may be linear, branched, or cyclic and is preferably linear or branched.

(75) The alkyl group represented by R^{sup.1} and R^{sup.2} may be unsubstituted or may have a substituent. Examples of the substituent include an aryl group, a heterocyclic group, a nitro group, a cyano group, a halogen atom, —OR^{sup.a}, —SR^{sup.a}, —COR^{sup.a}, —COOR^{sup.a}, —OCOR^{sup.a}, —NR^{sup.a}R^{sup.b}, —NHCOR^{sup.a}, —CONR^{sup.a}R^{sup.b}, —NHCONR^{sup.a}R^{sup.b}, —NHCOOR^{sup.a}, —SO₂R^{sup.a}, —SO₂OR^{sup.a}, and —NHSO₂R^{sup.a}. R^{sup.a} and R^{sup.b} each independently represent a hydrogen atom, an alkyl group, an aryl group, or a heterocyclic group. The substituent is preferably an aryl group. In particular, one of R^{sup.1} and R^{sup.2} is preferably an unsubstituted alkyl group while the other is preferably an alkyl group substituted with an aryl group.

(76) Examples of the halogen atom include a fluorine atom, a chlorine atom, a bromine atom, and an iodine atom.

(77) The alkyl group represented by R^{sup.a} and R^{sup.b} preferably has 1 to 20 carbon atoms. The alkyl group may be linear, branched, or cyclic and is preferably linear or branched.

(78) The number of carbon atoms in the aryl group as a substituent and the aryl group represented by R^{sup.a} and R^{sup.b} is preferably 6 to 20, more preferably 6 to 15, and even more preferably 6 to 10. The aryl group may be either a monocyclic ring or a condensed ring. In the aryl group, a part or all of the hydrogen atom is optionally substituted with an alkyl group having 1 to 8 carbon atoms.

(79) The heterocyclic group represented by R.sup.a and R.sup.b is preferably a 5-membered ring or a 6-membered ring. The heterocyclic group may be either a monocyclic ring or a condensed ring. The number of carbon atoms forming the heterocyclic group is preferably 3 to 30, more preferably 3 to 18, and more preferably 3 to 12. The number of hetero atoms forming the heterocyclic group is preferably 1 to 3. The hetero atom forming the heterocyclic group is preferably an nitrogen atom, oxygen atom, or sulfur atom.

(80) R.sup.3 and R.sup.4 each independently represent an hydrogen atom, an alkyl group having 1 to 12 carbon atoms, and an alkyl group having 2 to 4 carbon atoms substituted with an alkoxy group having 1 to 4 carbon atoms, or an alkenyl group having 3 to 5 carbon atoms. R.sup.3 and R.sup.4 may bond with each other to form an alkylene group having 3 to 7 carbon atoms, and the alkylene group may contain —O— or —N(R.sup.8)— in an alkylene chain thereof. R.sup.8 represents an alkyl group having 1 to 4 carbon atoms.

(81) R.sup.6 and R.sup.7 each independently represent an hydrogen atom, an alkyl group having 1 to 12 carbon atoms, and an alkyl group having 2 to 4 carbon atoms substituted with an alkoxy group having 1 to 4 carbon atoms, or an alkenyl group having 3 to 5 carbon atoms. R.sup.6 and R.sup.7 may bond with each other to form an alkylene group having 3 to 7 carbon atoms, and the alkylene group may contain —O— or —N(R.sup.9)— in an alkylene chain thereof. Herein, R.sup.9 represents an alkyl group having 1 to 4 carbon atoms.

(82) Examples of the photopolymerization initiator of the present invention (second photopolymerization initiator) include 2-methyl-1-[4-(methylthio)phenyl]-2-morpholinopropan-1-one (calculated pKa: 5.56), 2-benzyl-2-dimethylamino-1-(4-morpholinophenyl)-butanone-1 (calculated pKa: 7.68), 2-methyl-1-phenyl-2-morpholinopropan-1-one (calculated pKa: 5.56), 2-methyl-1-[4-(hexyl)phenyl]-2-morpholinopropan-1-one (calculated pKa: 5.56), 2-ethyl-2-dimethylamino-1-(4-morpholinophenyl)-butanone-1 (calculated pKa: 7.96), and 2-(dimethylamino)-2-[(4-methylphenyl)methyl]-1-[4-(4-morpholinyl)phenyl]-1-butanone (calculated pKa: 7.70).

(83) Examples of the photopolymerization initiator of the present invention (second photopolymerization initiator) commercially available include Omnirad 907, Omnirad 369E, and Omnirad 379 (trade name, all of which are manufactured by IGM Resins B.V.).

(84) The content of the photopolymerization initiator of the present invention (second photopolymerization initiator) contained in the pressure-sensitive adhesive layer of the present invention can be appropriately set as long as it can be cured by irradiation with radiation to provide excellent adhesive reliability, and is not particularly limited, but can be selected from the range of, for example, in terms of the upper limit, 2% by weight or less (e.g., 0.0001 to 2% by weight, 0.0001 to 1.8% by weight, 0.0001 to 1.6% by weight, 0.0001 to 1.4% by weight, 0.0001 to 1.2% by weight, 0.0001 to 1.0% by weight, 0.0001 to 0.8% by weight, 0.0001 to 0.6% by weight, 0.0001 to 0.4% by weight, and 0.0001 to 0.2% by weight); or in terms of the lower limit, 0.0001% by weight or more (e.g., 0.0001 to 2% by weight, 0.0002 to 2% by weight, 0.0003 to 2% by weight, 0.0004 to 2% by weight, 0.0005 to 2% by weight, 0.0006 to 2% by weight, 0.0007 to 2% by weight, 0.0008 to 2% by weight, 0.0009 to 2% by weight, 0.001 to 2% by weight, 0.0012 to 2% by weight, 0.0014 to 2% by weight, 0.0016 to 2% by weight, 0.0018 to 2% by weight, and 0.002 to 2% by weight) relative to the pressure-sensitive adhesive layer of the present invention (100% by weight).

(85) The pressure-sensitive adhesive layer of the present invention preferably further contains a hindered amine light stabilizer (HALS) in addition to the photopolymerization initiator of the present invention (second photopolymerization initiator) and the crosslinking agent of the present invention (second crosslinking agent). The configuration in which the pressure-sensitive adhesive layer of the present invention further contains the hindered amine light stabilizer is preferable in that the storage stability of the pressure-sensitive adhesive layer is improved, and excellent level difference absorbability and stress relaxation properties can be maintained for a long period of time. The hindered amine light stabilizer includes those of low and high molecular weights or N-

alkyl and N—H types. The hindered amine light stabilizer may be used alone or in a mixture of two or more.

(86) Examples of the hindered amine light stabilizer of low molecular weight include a hindered amine light stabilizer composed of 70% by weight of a reaction product (molecular weight: 737) of decanedioic acid bis[2,2,6,6-tetramethyl-1(octyloxy)-4-piperidinyl] ester, 1,1-dimethylethyl hydroperoxide, and octane and 30% by weight of polypropylene; bis(1,2,2,6,6-pentamethyl-4-piperidyl) [[3, 5-bis(1,1-dimethylethyl)-4-hydroxyphenyl]methyl]butylmalonate (molecular weight: 685); a mixture of bis(1,2,2,6,6-pentamethyl-4-piperidyl) sebacate and methyl-1,2,2,6,6-pentamethyl-4-piperidyl sebacate (molecular weight: 509); bis (2,2,6,6-tetramethyl-4-piperidyl) sebacate (molecular weight: 481); tetrakis (2,2,6,6-tetramethyl-4-piperidyl)-1,2,3,4-butanetetracarboxylate (molecular weight: 791); tetrakis (1,2,2,6,6-pentamethyl-4-piperidyl)-1,2,3,4-butanetetracarboxylate (molecular weight: 847); a mixture of 2,2,6,6-tetramethyl-4-piperidyl-1,2,3,4-butanetetracarboxylate and tridecyl 1,2,3,4-butanetetracarboxylate (molecular weight: 900); and a mixture of 1,2,2,6,6-pentamethyl-4-piperidyl-1,2,3,4-butanetetracarboxylate and tridecyl 1,2,3,4-butanetetracarboxylate (molecular weight: 900).

(87) Examples of the hindered amine light stabilizer of high molecular weight include poly[{6-(1,1,3,3-tetramethylbutyl)amino-1,3,5-triazine-2,4-diyl}{(2,2,6,6-tetramethyl-4-piperidyl)imino}hexamethylene{(2,2,6,6-tetramethyl-4-piperidyl)imino}] (molecular weight: 2000 to 3100); a polymer of dimethyl succinate and 4-hydroxy-2,2,6,6-tetramethyl-1-piperidineethanol (molecular weight: 3100 to 4000); a mixture of N,N',N'',N'''-tetrakis-(4,6-bis-(butyl-(N-methyl-2,2,6,6-tetramethylpiperidin-4-yl)amino)-triazin-2-yl)-4,7-diazadecane-1,10-diamine (molecular weight: 2286) and a polymer of dimethyl succinate and 4-hydroxy-2,2,6,6-tetramethyl-1-piperidineethanol; and a polycondensate of dibutylamine, 1,3,5-triazine, N,N'-bis(2,2,6,6-tetramethyl-4-piperidyl)-1,6-hexamethylenediamine, and N-(2,2,6,6-tetramethyl-4-piperidyl)butylamine (molecular weight: 2600 to 3400).

(88) When, the pressure-sensitive adhesive layer of the present invention contains the hindered amine light stabilizer, the hindered amine light stabilizer to be used preferably has a pKa of 5 to 12.7. The configuration in which the hindered amine light stabilizer has a pKa of 5 to 12.7 is suitable in terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer of the present invention during storage of the radiation curable pressure-sensitive adhesive sheet of the present invention, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability.

(89) The pKa (acid dissociation constant) of the hindered amine light stabilizer is an index for quantitatively indicating the strength of an acid, and the smaller the pKa, the stronger the acidity. It is considered that when the pressure-sensitive adhesive has a high acid value, the second crosslinking agent (especially, polyfunctional (meth)acrylate) undergoes a Michael addition reaction, resulting in a gradual crosslinking reaction. When the pKa of the hindered amine light stabilizer of the present invention is in the range of 5 to 12.7, it is presumed that the acid value in the pressure-sensitive adhesive layer is adjusted to suppress the crosslinking reaction over time. Note that this mechanism is presumptive and should not be construed as limiting the scope of the present invention.

(90) In the radiation curable pressure-sensitive adhesive sheet of the present invention, the pKa of the hindered amine light stabilizer is preferably 5.1 or more and more preferably 5.2 or more, and may be 5.3 or more or 5.4 or more, in terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer during storage, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability. In terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer during storage, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability, the pKa of the hindered amine light stabilizer is also preferably 12.5 or less and more preferably 12 or less, and may be 11.5 or

less, 11 or less, 10.5 or less, 10 or less, or 9.5 or less.

(91) As the pKa of the hindered amine light stabilizer, it is preferable to use a calculated pKa as in the case of the photopolymerization initiator of the present invention. Specifically, a chemical structural formula of the hindered amine light stabilizer is drawn using ChemDraw 19.0, and the chemical structural formula with a calculated pKa of 5 to 12.7 obtained using a computational module is preferably used.

(92) When the pressure-sensitive adhesive layer of the present invention contains a hindered amine light stabilizer, the content thereof can be appropriately set as long as it can be cured by irradiation with radiation to provide excellent adhesive reliability, and is not particularly limited, but can be selected from the range of, for example, in terms of the upper limit, 8% by weight or less (e.g., 0.001 to 8% by weight, 0.001 to 7.5% by weight, 0.001 to 7.0% by weight, 0.001 to 6.5% by weight, 0.001 to 6.0% by weight, 0.001 to 5.5% by weight, 0.001 to 5.0% by weight, 0.001 to 4.5% by weight, and 0.001 to 4.0% by weight); or in terms of the lower limit, 0.001% or more (e.g., 0.001 to 8% by weight, 0.002 to 8% by weight, 0.003 to 8% by weight, 0.004 to 8% by weight, 0.005 to 8% by weight, 0.006 to 8% by weight, 0.007 to 8% by weight, 0.008 to 8% by weight, 0.009 to 8% by weight, and 0.01 to 8% by weight) relative to the pressure-sensitive adhesive layer of the present invention (100% by weight), in terms of being able to suppress the progress of curing of the pressure-sensitive adhesive layer during storage, being able to maintain excellent level difference absorbability and stress relaxation properties for a long time, and having excellent storage stability.

(93) The method of forming the pressure-sensitive adhesive layer of the present invention is not particularly limited, and examples of one embodiment (sometimes referred to herein as “first embodiment”) include applying (coating) a pressure-sensitive adhesive composition containing the base pressure-sensitive adhesive material, the photopolymerization initiator of the present invention (second photopolymerization initiator), and the crosslinking agent of the present invention (second crosslinking agent), if necessary, a hindered amine light stabilizer (HALS), and other additives on a support to dry and cure the resulting pressure-sensitive adhesive composition layer; and applying (coating) the pressure-sensitive adhesive composition on a support to cure the resulting pressure-sensitive adhesive composition layer by irradiation with an active energy ray. If necessary, the methods may further include heating and drying.

(94) In order to prevent the crosslinking reaction of the photopolymerization initiator of the present invention and the crosslinking agent of the present invention from proceeding by irradiation with an active energy ray, the pressure-sensitive adhesive composition layer is preferably dried and cured.

(95) The support is not particularly limited and is preferably a plastic film. Examples of materials of the plastic film and the like include plastic materials such as polyester resins including polyethylene terephthalate (PET); acrylic resins including polymethyl methacrylate (PMMA); polycarbonate; triacetyl cellulose (TAC); polysulfone; polyacrylate; polyimide; polyvinyl chloride; polyvinyl acetate; polyethylene; polypropylene; ethylene-propylene copolymer; cyclic olefin-based polymer including trade name “ARTON” (cyclic olefin-based polymer, manufactured by JSR Corporation) and trade name “ZEONOR” (cyclic olefin-based polymer, Zeon Corporation). Note that these plastic materials may be used either alone or in combination of two or more thereof.

(96) The support may be a release sheet. Examples of the release sheet include, but not particularly limited to, a plastic film whose surface is treated by a release agent such as silicon type, long-chain alkyl type, fluorine type, and molybdenum sulfide.

(97) For the application (coating) in the pressure-sensitive adhesive composition, it is possible to use known coating processes, and examples thereof include a coater, such as a gravure roll coater a reverse roll coater, a kiss roll coater, a dip roll coater, a bar coater, a knife coater, a spray coater, a comma coater, or a direct coater.

(98) The drying and curing temperature is preferably 40 to 200° C., more preferably 50 to 180° C.,

and even more preferably 60 to 170° C. The drying and curing time may be adopted as appropriate and is, for example, 5 seconds to 20 minutes, preferably 5 seconds to 10 minutes, and more preferably 10 seconds to 5 minutes.

(99) Examples of the active energy ray include ionizing radiations such as an α -ray, a β -ray, a γ -ray, a neutron ray, and an electron ray and ultraviolet rays, and particularly, the ultraviolet rays are preferable. Irradiation energy, irradiation time, and irradiation method of the active energy ray are not particularly limited, and each may be set appropriately to achieve a desired viscosity and viscoelasticity, depending on a thickness of a pressure-sensitive adhesive layer or the like.

(100) A principal surface of the pressure-sensitive adhesive layer formed above that does not face the support is preferably further laminated with another support (including a release sheet) to block oxygen that inhibits light curing when the pressure-sensitive adhesive layer is light-cured by the above active energy ray and/or ultraviolet irradiation described later, etc.

(101) Another embodiment of the method of forming the pressure-sensitive adhesive layer according to the present invention (sometimes referred to herein as “second embodiment”) preferably involves, for example, forming it by a method including the following processes: (a) forming a pressure-sensitive adhesive layer formed of a base pressure-sensitive adhesive material on a support (pressure-sensitive adhesive layer forming step); (b) curing the pressure-sensitive adhesive layer (pressure-sensitive adhesive layer curing step); (c) providing a solution of the photopolymerization initiator of the present invention and the crosslinking agent of the present invention to apply the solution to one of opposite surfaces of the cured pressure-sensitive adhesive layer (solution application step); (d) causing the photopolymerization initiator of the present invention and the crosslinking agent of the present invention contained in the solution to infiltrate from the one surface in a thickness direction of the pressure-sensitive adhesive layer (solution infiltration step); and (e) drying the pressure-sensitive adhesive layer (drying step).

(102) In the method of the first embodiment, when the pressure-sensitive adhesive composition contains the first photopolymerization initiator for forming the crosslinked structure of the base pressure-sensitive adhesive material and the photopolymerization initiator of the present invention (second photopolymerization initiator), the light absorption wavelength bands of the two photopolymerization initiators need to be sufficiently separated from each other and to distinguish the wavelength regions so that only the curing reaction by the first photopolymerization initiator proceeds while the curing reaction by the second photopolymerization initiator does not proceed, resulting in extremely limited design possibilities. Even when a thermal polymerization initiator is used as the polymerization initiator for forming the crosslinked structure of the base pressure-sensitive adhesive material, curing may proceed by the reaction of the second photopolymerization initiator during thermal curing. Thus, strict control of the curing reaction is required. There is also the problem that thick pressure-sensitive adhesive layers take longer to thermally cure, resulting in lower production efficiency.

(103) On the other hand, since the curing reaction by the first photopolymerization initiator is completed before the addition of the second photopolymerization initiator in the method of the second embodiment, it is not necessary to combine the first photopolymerization initiator or set strict curing conditions that would prevent the curing reaction by the second photopolymerization initiator from progressing, and thus the degree of freedom in designing a pressure-sensitive adhesive is dramatically improved. In other words, since curing reactions of the first and second photopolymerization initiators are separated, the degree of freedom in selecting the combination of the first and second photopolymerization initiator is extremely wide, and thus light absorption wavelength bands of the two photopolymerization initiators can overlap or approximate each other. Furthermore, it is also possible to use a combination of the same polymerization initiator as the first and second polymerization initiators, which had been impossible in the past.

(104) Thus, the extremely high degree of freedom in the combination of the first and second photopolymerization initiators and the possibility of combining identical photopolymerization

initiators are also suitable for avoiding the use of polymerization initiators that may cause coloration and degradation of the pressure-sensitive adhesive layer. Moreover, since the curing reactions of the first and second photopolymerization initiators are separated and do not interfere with each other, there is no need to set strict conditions to control the respective curing reactions. For example, it is not necessary to set conditions for the first photopolymerization initiator that prevent the second photopolymerization initiator from progressing. Once the curing conditions capable of imparting excellent level difference absorbability and stress relaxation properties to the pressure-sensitive adhesive layer are determined, the adhesive reliability to be imparted to the pressure-sensitive adhesive layer can be easily controlled by changing the coating and curing conditions for the solution of the second photopolymerization initiator afterward. In addition, even in the case of the combination of the thermal polymerization initiator and the second photopolymerization initiator, since the two curing reactions are separated from each other, there is no need to control to suppress the curing by the second photopolymerization initiator during thermal curing.

(105) FIGS. 2 (a) to (e) are views schematically illustrating a process for implementing one embodiment of the production method according to the second embodiment of the present invention.

(106) As illustrated in FIG. 2(a), a pressure-sensitive adhesive layer **10a** formed of a transparent base pressure-sensitive adhesive material is first formed on a support **S1** (pressure-sensitive adhesive layer forming step). As the support, one illustrated as the support of the first embodiment can be used.

(107) In the second embodiment, the method of forming the pressure-sensitive adhesive layer is not particularly limited, and the same method as in the above first embodiment can be used.

(108) Next, the pressure-sensitive adhesive layer **10a** is cured (curing step of pressure-sensitive adhesive layer). In FIG. 2(b), **10** is a pressure-sensitive adhesive layer being the cured pressure-sensitive adhesive layer **10a**. Examples of methods of curing the pressure-sensitive adhesive layer **10a** include, but not particularly limited to, heating the pressure-sensitive adhesive layer **10a** and curing the pressure-sensitive adhesive layer **10a** by irradiation with the active energy ray. If necessary, the methods may further include heating and drying. Examples of the active energy ray include ionizing radiations such as an x-ray, a β -ray, a γ -ray, a neutron ray, and an electron ray and ultraviolet rays, and particularly, the ultraviolet rays are preferable.

(109) Conditions of curing the pressure-sensitive adhesive layer **10a** can be appropriately selected so that the pressure-sensitive adhesive layer **10** has desired physical properties by an embodiment. For example, in the radiation curable pressure-sensitive adhesive sheet of the present invention, heating temperature and time or an irradiation dose of the active energy rays may be set appropriately so that the pressure-sensitive adhesive layer **10** exhibits high fluidity and excellent level difference absorbability and stress relaxation property.

(110) FIG. 2(b) is an embodiment in which the pressure-sensitive adhesive layer **10a** is cured by irradiating the pressure-sensitive adhesive layer **10a** with an ultraviolet ray **U**. The pressure-sensitive adhesive layer **10a** may be directly irradiated with ultraviolet rays but is preferably irradiated through a support to block oxygen that inhibits curing by ultraviolet irradiation. FIG. 2(b) is an embodiment in which the pressure-sensitive adhesive layer **10a** is irradiated with ultraviolet rays through a support **S2**. When ultraviolet rays are irradiated through a support, another support **S2** (including a release sheet) is laminated onto a principal surface of the pressure-sensitive adhesive layer **10a** opposite to the principal surface facing the support **S1** to irradiate the ultraviolet rays through the support. The illuminance and time of ultraviolet irradiation are appropriately set according to the composition of the base pressure-sensitive adhesive material and the thickness of the pressure-sensitive adhesive layer. For the ultraviolet irradiation, a high-pressure mercury lamp, a low-pressure mercury lamp, a metal halide lamp, and the like can be used.

(111) Next, as illustrated in FIG. 2(c), after the support **S2** is released and removed, a solution **12** of

a photopolymerization initiator **11a** (second photopolymerization initiator) and a crosslinking agent **11b** (second crosslinking agent) is applied to one of opposite surfaces of the pressure-sensitive adhesive layer **10** (solution application step). The solution **12** of the photopolymerization initiator **11a** and the crosslinking agent **11b** is not particularly limited as long as it can be coated on the pressure-sensitive adhesive layer in liquid form and infiltrates. When the photopolymerization initiator **11a** and/or the crosslinking agent **11b** are in liquid form, for example, each may be applied as a solution as it is, or the solution may be obtained by mixing the photopolymerization initiator **11a** and the crosslinking agent **11b**. It may also be a solution in which the photopolymerization initiator **11a** and the crosslinking agent **11b** are dissolved in a solvent. FIG. 2(c) is an embodiment of applying a solution **12** in which a photopolymerization initiator **11a** and a crosslinking agent **11b** are dissolved in a solvent **13** to one of opposite surfaces of the pressure-sensitive adhesive layer **10**. (112) On the surface of the pressure-sensitive adhesive layer **10**, the photopolymerization initiator **11a** and the crosslinking agent **11b** in the solution **12** infiltrate into the pressure-sensitive adhesive layer **10** in the thickness direction (solution infiltration step). This state is illustrated in FIG. 2(d). When the solution **12** is a solution in which the photopolymerization initiator **11a** and the crosslinking agent **11b** are dissolved in the solvent **13**, the surface of the pressure-sensitive adhesive layer **10a** is swollen by the infiltration of the solvent **13**, and the photopolymerization initiator **11a** and the crosslinking agent **11b** infiltrate into the pressure-sensitive adhesive layer **10** in a dissolved state in the solvent. The photopolymerization initiator **11a** and the crosslinking agent **11b** become “dissolved” within the pressure-sensitive adhesive layer **10**.

(113) In the process of causing the photopolymerization initiator **11a** and the crosslinking agent **11b** to infiltrate the pressure-sensitive adhesive layer **10a**, a concentration gradient can be formed in the thickness direction. Therefore, the concentration of the photopolymerization initiator **11a** and the crosslinking agent **11b** on a side to which the solution **12** is applied can be higher than on the opposite side. This state is illustrated in FIG. 2(d).

(114) Thereafter, a pressure-sensitive adhesive sheet **1a** as illustrated in FIG. 2(e) can be obtained by drying the pressure-sensitive adhesive layer **10** (drying step). When the solution **12** is a solution in which the photopolymerization initiator **11a** and the crosslinking agent **11b** are dissolved in the solvent **13**, the infiltrated solvent **13** is evaporated by drying step. The pressure-sensitive adhesive layer **10** returns to a state close to that before application by drying the pressure-sensitive adhesive layer **10**. Therefore, a change in the physical properties of the pressure-sensitive adhesive layer **10** due to the photopolymerization initiator **11a** and the crosslinking agent **11b** can be minimized.

Once the pressure-sensitive adhesive layer **10** is dried, the infiltration of the photopolymerization initiator **11a** and the crosslinking agent **11b** into the pressure-sensitive adhesive layer **10** stops, and the concentration gradient of the photopolymerization initiator **11a** and the crosslinking agent **11b** is fixed.

(115) The pressure-sensitive adhesive layer **10a** contains a first photopolymerization initiator and a first crosslinking agent; the pressure-sensitive adhesive layer curing step is preferably curing by reaction of the first photopolymerization initiator and the first crosslinking agent. When the pressure-sensitive adhesive layer **10a** contains the first photopolymerization initiator and the first crosslinking agent, after the pressure-sensitive adhesive layer curing step, both or either one of the first photopolymerization initiator and the first crosslinking agent may remain. In this case, the solution **12** may contain only one of the second photopolymerization initiator and the second crosslinking agent. However, it is preferable that the solution **12** contains both the second photopolymerization initiator and the second crosslinking agent.

(116) The solvent is not particularly limited as long as it can dissolve the second photopolymerization initiator and the second crosslinking agent and swell the pressure-sensitive adhesive layer **10**, but a non-aqueous solvent is preferred since an aqueous solvent has poor wettability to the pressure-sensitive adhesive layer and the photopolymerization initiator and the crosslinking agent does not easily infiltrate. Examples of the non-aqueous solvent include, but not

particularly limited to, esters such as methyl acetate, ethyl acetate, isopropyl acetate, and butyl acetate; aromatic hydrocarbons such as toluene, xylene, and ethylbenzene; ketones such as acetone, methyl ethyl ketone, and methyl isobutyl ketone; alicyclic ketones such as cyclopentanone and cyclohexanone; aliphatic hydrocarbons such as hexane, heptane, and octane; alicyclic hydrocarbons such as cyclohexane; halogenated hydrocarbons such as chloroform, dichloromethane, and 1,2-dichloroethane; ethers such as diethyl ether, dimethoxyethane, tetrahydrofuran, and dioxane; amides such as N,N-dimethylformamide and N,N-dimethylacetamide; nitriles such as acetonitrile, propionitrile, and benzonitrile; and alcohols such as methanol, ethanol, propanol, isopropanol, n-butanol, isobutanol, sec-butanol, and tert-butanol, and esters, aromatic hydrocarbons, ketones, and alcohols are preferable. The solvent can also be used alone or in combination of two or more.

(117) The total concentration of the second photopolymerization initiator and the second crosslinking agent in the solution can be appropriately set according to the desired level difference absorbability, stress relaxation property, and adhesive reliability to be imparted to the pressure-sensitive adhesive layer **10** and for example, can be appropriately selected from the range of 95% by weight or less (e.g., 0.1 to 95% by weight, 0.1 to 90% by weight, 0.1 to 85% by weight, 0.1 to 80% by weight, 0.1 to 70% by weight, 0.1 to 60% by weight, 0.1 to 50% by weight, and 0.1 to 40% by weight) and 0.1% by weight or more (e.g., 0.1 to 95% by weight, 0.2 to 95% by weight, 0.3 to 95% by weight, 0.4 to 95% by weight, 0.5 to 95% by weight, 1 to 95% by weight, 1.5 to 95% by weight, 3 to 95% by weight, and 5 to 95% by weight). If the concentration of the second photopolymerization initiator and the second crosslinking agent in the solution is within this range, the second photopolymerization initiator and the second crosslinking agent can be dissolved, the pressure-sensitive adhesive layer **10** can be sufficiently swollen, and excellent level difference absorbability, stress relaxation property, and adhesive reliability may be imparted to the pressure-sensitive adhesive layer **10**.

(118) Specifically, the concentration of the second crosslinking agent in solution can be appropriately set according to the desired curing properties to be imparted to the pressure-sensitive adhesive layer **10** and for example, can be appropriately selected from the range of 95% by weight or less (e.g., 1 to 95% by weight, 1 to 90% by weight, 1 to 85% by weight, 1 to 80% by weight, 1 to 75% by weight, 1 to 70% by weight, 1 to 65% by weight, and 1 to 60% by weight) and 1% by weight or more (e.g., 1 to 95% by weight, 3 to 95% by weight, 4 to 95% by weight, 5 to 95% by weight, 6 to 95% by weight, 7 to 95% by weight, 8 to 95% by weight, 9 to 95% by weight, and 10 to 95% by weight).

(119) In addition, the concentration of the second photopolymerization initiator in the solution can be appropriately set according to the desired curing properties to be imparted to the pressure-sensitive adhesive layer **10**, and can be selected from the range of, for example, in terms of the upper limit of 20% by weight or less (e.g., 0.001 to 20% by weight, 0.001 to 19% by weight, 0.001 to 18% by weight, 0.001 to 17% by weight, 0.001 to 16% by weight, 0.001 to 15% by weight, 0.001 to 14% by weight, 0.001 to 13% by weight, and 0.001 to 12% by weight); or in terms of the lower limit, 0.001% or more (e.g., 0.001 to 20% by weight, 0.002 to 20% by weight, 0.003 to 20% by weight, 0.004 to 20% by weight, 0.005 to 20% by weight, 0.006 to 20% by weight, 0.007 to 20% by weight, 0.008 to 20% by weight, 0.009 to 20% by weight, and 0.01 to 20% by weight).

(120) When the pressure-sensitive adhesive layer of the present invention contains a hindered amine light stabilizer (HALS), another hindered amine light stabilizer may be dissolved in the solution **12**. When the solution **12** contains an hindered amine light stabilizer in addition to the second photopolymerization initiator and the second crosslinking agent, the pressure-sensitive adhesive layer of the present invention containing the hindered amine light stabilizer can be produced in a single coating, thereby improving production efficiency. The concentration of the hindered amine light stabilizer in the solution can be appropriately set according to the desired storage stability to be imparted to the pressure-sensitive adhesive layer **10**, and can be selected from the range of, for example, in terms of the upper limit of 20% by weight or less (e.g., 0.001 to

20% by weight, 0.001 to 19% by weight, 0.001 to 18% by weight, 0.001 to 17% by weight, 0.001 to 16% by weight, 0.001 to 15% by weight, 0.001 to 14% by weight, 0.001 to 13% by weight, and 0.001 to 12% by weight); or in terms of the lower limit, 0.001% or more (e.g., 0.001 to 20% by weight, 0.002 to 20% by weight, 0.003 to 20% by weight, 0.004 to 20% by weight, 0.005 to 20% by weight, 0.006 to 20% by weight, 0.007 to 20% by weight, 0.008 to 20% by weight, 0.009 to 20% by weight, and 0.01 to 20% by weight).

(121) If the concentration of the above second photopolymerization initiator, second crosslinking agent, and hindered amine light stabilizer are each higher than the above range, they may be bled out and, distribution variations may occur from the viewpoint of application uniformity. If the concentration is lower than the above range, more solvent than necessary may be required, resulting in a decrease in adhesive properties due to residual solvent and appearance defects (uneven surface) due to swelling of the pressure-sensitive adhesive more than necessary.

(122) The solution **12** also preferably contains an ultraviolet absorbing agent. When the solution **12** contains an ultraviolet absorbing agent in addition to the second photopolymerization initiator and the second crosslinking agent, a pressure-sensitive adhesive sheet containing the ultraviolet absorbing agent can be produced in a single coating, thereby improving production efficiency. The solution **12** is even more preferable because if a pressure-sensitive adhesive sheet of the present invention containing an ultraviolet absorbing agent is irradiated with ultraviolet rays, the ultraviolet absorbing agent that has absorbed the ultraviolet rays generates heat, which accelerates the curing reaction and thus improves adhesive reliability. The concentration of the ultraviolet absorbing agent in the solution can be appropriately set according to the desired ultraviolet absorption properties to be imparted to the pressure-sensitive adhesive layer **10** and for example, can be selected from the range of, in terms of the upper limit, 50% by weight or less (e.g., 1 to 50% by weight, 1 to 45% by weight, 1 to 40% by weight, 1 to 35% by weight, 1 to 30% by weight, 1 to 25% by weight, 1 to 20% by weight, and 1 to 15% by weight); or in terms of the lower limit, 1% or more (e.g., 1 to 50% by weight, 2 to 50% by weight, 3 to 50% by weight, 4 to 50% by weight, and 5 to 50% by weight).

(123) For the application (coating) to the pressure-sensitive adhesive layer **10** in the solution **12**, it is possible to use known coating processes, and examples thereof include a coater, such as a gravure roll coater a reverse roll coater, a kiss roll coater, a dip roll coater, a bar coater, a knife coater, a spray coater, a comma coater, or a direct coater.

(124) The amount of the solution **12** applied to the pressure-sensitive adhesive layer **10** can be appropriately set according to the desired level difference absorbability, stress relaxation properties, and adhesive reliability to be imparted to the pressure-sensitive adhesive layer **10** and for example, can be selected from the range of 1 to 1000 $\mu\text{g}/\text{cm}^2$, preferably 1 to 500 $\mu\text{g}/\text{cm}^2$, more preferably 1 to 300 $\mu\text{g}/\text{cm}^2$, and even more preferably 1 to 100 $\mu\text{g}/\text{cm}^2$. If the amount of the solution **12** applied is within this range, the second photopolymerization initiator and the second crosslinking agent can be dissolved, and the level difference absorbability, stress relaxation properties, and adhesive reliability may be sufficiently imparted to the pressure-sensitive adhesive layer **10**.

(125) After the solution **12** is applied to the pressure-sensitive adhesive layer **10**, it may be allowed to stand to cause the second photopolymerization initiator and the second crosslinking agent to infiltrate, if necessary. The standing time is not particularly limited, and can be appropriately selected, for example, within 15 minutes and can be selected from the range of, for example, 1 second to 10 minutes and preferably 5 seconds to 5 minutes. The standing temperature can be room temperature (approximately 10 to 30° C.). When allowed to stand under the above conditions, the solution can sufficiently infiltrate the pressure-sensitive adhesive layer **10**.

(126) A heat-drying temperature in the drying step is preferably 40 to 200° C., more preferably 50 to 180° C., and even more preferably 60 to 170° C. The drying time may be adopted as appropriate and is, for example, 5 seconds to 20 minutes, preferably 5 seconds to 10 minutes, and more preferably 10 seconds to 5 minutes. The pressure-sensitive adhesive layer **10** can be returned to a

state close to that before application by drying it under the above conditions.

(127) If necessary, the standing time may be designed to make the second photopolymerization initiator and the second crosslinking agent and the pressure-sensitive adhesive layer more uniform. The standing time is not particularly limited, and can be appropriately selected, for example, within 30 days, and can be appropriately selected, for example, between 1 hour and 15 days and preferably between 24 hours and 10 days. The second photopolymerization initiator and the second crosslinking agent and the pressure-sensitive adhesive layer are stabilized in the pressure-sensitive adhesive layer **10** by allowing it to stand, thereby suppressing variation in characteristic evaluation.

(128) FIG. **3** is a cross-sectional view illustrating one embodiment of the radiation curable pressure-sensitive adhesive sheet according to the present invention produced by the production method of the above second embodiment.

(129) A radiation curable pressure-sensitive adhesive sheet **1b** according to one embodiment of the present invention in the FIG. **3** is composed of: a pressure-sensitive adhesive layer **10**, a release sheet **S1** being laminated onto the other principal surface **10A** of the pressure-sensitive adhesive layer **10**, and a release sheet **S2** being laminated onto opposite principal surface **10B** of the pressure-sensitive adhesive layer **10**.

(130) In FIG. **3**, the pressure-sensitive adhesive layer **10** has two opposite principal surfaces (first principal surface **10A** and second principal surface **10B**) comprising a base pressure-sensitive adhesive material and may be a single layer or a laminated structure of two or more layers. The pressure-sensitive adhesive layer **10** may be formed through the pressure-sensitive adhesive layer forming step and the pressure-sensitive adhesive layer curing step and is equivalent to the pressure-sensitive adhesive layer **10** in FIG. **2** described above. Therefore, the pressure-sensitive adhesive layer **10** is preferably a cured pressure-sensitive adhesive layer.

(131) The “single layer” of the pressure-sensitive adhesive layer means that it does not have a laminated structure. For example, a formation of a pressure-sensitive adhesive layer comprising a transparent base pressure-sensitive adhesive material with a pressure-sensitive adhesive layer comprising the same transparent base pressure-sensitive adhesive material formed thereon has a laminated structure, not a single layer.

(132) The thickness of the pressure-sensitive adhesive layer **10** is not particularly limited but is typically 5 μm to 500 μm , preferably 5 μm to 400 μm , even more preferably 5 μm to 350 μm . If the thickness of the pressure-sensitive adhesive layer **10** is in this range, the pressure-sensitive adhesive layer **10** can follow such that no bubbles are left in the stepped space and reduce stress strain caused by the stepped space to suppress display unevenness, and has excellent level difference absorbability and stress relaxation properties, which is preferable.

(133) As measured according to JIS K7361, total light transmittance of the entire pressure-sensitive adhesive layer **10** is not particularly limited but is preferably 80% or more and preferably 90% or more. The higher total light transmittance of the pressure-sensitive adhesive layer **10** provides better results. Furthermore, a haze value thereof is preferably 1.5% or less and more preferably 1% or less and 0.8% or less.

(134) The pressure-sensitive adhesive layer **10** contains a photopolymerization initiator **11a** and a crosslinking agent **11b**. The photopolymerization initiator **11a** and the crosslinking agent **11b** are preferably dissolved in the pressure-sensitive adhesive layer **10**. Herein, “dissolution” means, for example, that the photopolymerization initiator **11a** and the crosslinking agent **11b** dissolve to such an extent that the transparency of the pressure-sensitive adhesive layer **10** can be maintained, i.e., white turbidity caused by light scattering does not occur. Specifically, the photopolymerization initiator **11a** and the crosslinking agent **11b** are preferably contained in the pressure-sensitive adhesive layer **10** so that the haze value thereof is 1.5% or less, preferably 1% or less, and 0.8% or less. The present invention encompasses an embodiment in which either one of the photopolymerization initiator **11a** and the crosslinking agent **11b** is dissolved in the pressure-sensitive adhesive layer **10**.

(135) In FIG. 3, the dotted line X-X' is a line dividing the pressure-sensitive adhesive layer **10** into two equal portions in a thickness direction. When the thickness of the pressure-sensitive adhesive layer **10** is not uniform, the dotted line X-X' is a line bisecting the thickness at each point.

(136) The photopolymerization initiator **11a** and the crosslinking agent **11b**, are formed by causing the photopolymerization initiator **11a** and the crosslinking agent **11b** to infiltrate the pressure-sensitive adhesive layer **10** through the solution application step, the solution infiltration step, and the drying step, and as illustrated in FIG. 3, a concentration gradient of the photopolymerization initiator **11a** and the crosslinking agent **11b** may occur in the thickness direction of the pressure-sensitive adhesive layer **10**. Therefore, in a case where the single pressure-sensitive adhesive layer **10** is divided into two equal portions in a thickness direction, the concentration of the photopolymerization initiator **11a** and the crosslinking agent **11b** in the area to which one of the two principal surfaces, the first principal surface **10A**, belongs is different from the concentration of the photopolymerization initiator **11a** and the crosslinking agent **11b** in the area to which the other principal surface, the second principal surface **10B**, belongs. The scope of the present invention also includes a case where the photopolymerization initiator **11a** and the crosslinking agent **11b** are not present in a region where the concentration is lower (the concentration is 0).

(137) The concentration of the photopolymerization initiator **11a** and the crosslinking agent **11b** in the area to which the first principal surface belongs and the concentration of the photopolymerization initiator **11a** and the crosslinking agent **11b** in the area to which the second principal surface belongs mean the average concentration in each region if there is also a concentration gradient in each region.

(138) FIG. 3 illustrates an embodiment in which the first principal surface **10A** faces the support **S1**, and the concentration of the photopolymerization initiator **11a** and the crosslinking agent **11b** in the area to which the second principal surface **10B** belongs is higher than the concentration of the photopolymerization initiator **11a** and the crosslinking agent **11b** in the area to which the first principal surface **10A** belongs, and this can be obtained by applying the solution **12** to the second principal surface **10B** to cause the photopolymerization initiator **11a** and the crosslinking agent **11b** into the pressure-sensitive adhesive layer **10** to infiltrate in a dissolved state from the first principal surface **10B** over a depth in the thickness direction.

(139) The radiation curable pressure-sensitive adhesive sheet according to the present invention can be used for bonding a transparent optical element to another optical element in an image display device, such as a liquid crystal image display device or an organic EL image display device. Examples of the optical elements include various types of transparent optical elements, such as a polarizing film, a retardation film, and a transparent cover element including a cover glass. The optical elements of the present invention may also include a glass substrate in which a transparent electroconductive layer such as a patterned ITO film is formed.

(140) FIG. 4 is a cross-sectional view of an optical element laminate presented as one example of the simplest embodiment using a radiation curable pressure-sensitive adhesive sheet of the present invention. Referring to FIG. 4, the optical element laminate **2** is composed of: an optically transparent, first optical element **S3**; and a second optical element **S4** bonded to the first optical element **S3** through an optically transparent pressure-sensitive adhesive layer **10**. The optical element laminate **2** is obtained by peeling off the supports **S1** and **S2** from the pressure-sensitive adhesive sheet **1b** illustrated in FIG. 3 to laminate them to the first and second optical elements. The transparent, first optical element **S3** and second optical element **S4** may be composed of: an optical film for use in an optical display device, such as a polarizing film or a retardation film; or a transparent cover element such as a viewing-side cover glass of an optical display device. The first optical element **S3** and the second optical element **S4** are respectively bonded to the first principal surface **10A** and the second principal surface **10B** of the pressure-sensitive adhesive layer **10**.

(141) FIGS. 5(a)-5(d) are views schematically illustrating a process for implementing one embodiment of a method for producing an optical element laminate using the radiation curable

pressure-sensitive adhesive sheet of the present invention. In the present embodiment, a substrate **20** comprising an optical element laminate **2a** and an optical element (hereinafter, sometimes referred to simply as “substrate **20**”) is used, as illustrated in FIG. 5(a).

(142) In the present embodiment, the optical element laminate **2a** is obtained by laminating the optical element **S4** onto the principal surface **10B** of the pressure-sensitive adhesive sheet **1a** produced by the production method according to the second embodiment of the present invention to remove the support **S1**.

(143) In the optical element laminate **2a** in FIG. 5(a), a pressure-sensitive adhesive layer **10** in the present embodiment is cured by the pressure-sensitive adhesive layer curing step, and a photopolymerization initiator **11a** and a crosslinking agent **11b** are dispersed in a dissolved state. The present invention encompasses an embodiment in which either one of the photopolymerization initiator **11a** and the crosslinking agent **11b** is dissolved in the pressure-sensitive adhesive layer **10**.

(144) In the present embodiment, the optical element laminate **2a** has an optical element **S4** but the optical element **S4** may be the support **S2** (release sheet). In the present embodiment, there is a concentration gradient of the photopolymerization initiator **11a** and the crosslinking agent **11b** in a thickness direction from a principal surface **10B** in which the pressure-sensitive adhesive layer **10** is in contact with the optical element **S4**.

(145) In a substrate **20** in FIG. 5(a), a principal surface **20a** bonded to the optical element laminate **2a** in the present embodiment have a printed layer **30**. The printed layer **30** includes a transparent electroconductive printed layer such as a patterned ITO (Indium Tin Oxide) and a black concealing portion formed in a frame shape in a peripheral edge portion of a transparent cover element. The present invention also encompasses the case where a substrate **20** without the printed layer **30** is used.

(146) Next, the pressure-sensitive adhesive layer **10** of the optical element laminate **2a** is bonded to the principal surface **20a** of the substrate **20**. The bonding can be performed by a known method, for example, under heating and pressurizing conditions using an autoclave. The pressure-sensitive adhesive layer **10** of the optical element laminate **2a** is cured by the pressure-sensitive adhesive layer curing step but is in a state before the curing reaction by the photopolymerization initiator **11a** (second photopolymerization initiator) and the crosslinking agent **11b** (second crosslinking agent) proceeds, so the pressure-sensitive adhesive layer **10** exhibits high fluidity and excellent level difference absorbability and stress relaxation property. Therefore, the pressure-sensitive adhesive layer **10** is bonded so as to fill up a stepped space between the principal surface **20A** of the substrate **20** and the printed layer **30**. In addition, the pressure-sensitive adhesive layer **10** can reduce sufficiently stress strain caused by the stepped space **30** and can suppress display unevenness when the optical element laminate is used in an image display device.

(147) Next, the pressure-sensitive adhesive layer **10** is cured by reaction of photopolymerization initiator **11a** and the crosslinking agent **11b**. Examples of the methods of curing the pressure-sensitive adhesive layer **10**, include curing the pressure-sensitive adhesive layer **10** by irradiation with the active energy ray. If necessary, the methods may further include heating and drying. Examples of the active energy ray include ionizing radiations such as an α -ray, a β -ray, a γ -ray, a neutron ray, an electron ray, and ultraviolet rays, and particularly, the ultraviolet rays are preferable.

(148) The conditions for curing the pressure-sensitive adhesive layer **10**, for example, the heating temperature and time or the irradiation dose of the active energy rays, may be set appropriately so as to exhibit high elastic modulus and excellent adhesive reliability.

(149) FIG. 5(c) is an embodiment in which the pressure-sensitive adhesive layer **10** is cured by irradiating the pressure-sensitive adhesive layer **10** with ultraviolet rays **U**. The irradiation with the ultraviolet rays **U** decomposes the photopolymerization initiator **11a** and generates radicals, ions, or the like to initiate polymerization and crosslinking reaction of the crosslinking agent **11b**. The ultraviolet rays may be irradiated through the optical element **S4**. When the optical element **S4** is a support **S2** (release sheet), the optical element **S4** may be laminated after ultraviolet rays are

irradiated through the support **S2** to release the support **S2**. Moreover, the ultraviolet rays may be irradiated from the substrate **20** (optical element) side. FIG. 5(c) is an embodiment in which the pressure-sensitive adhesive layer **10** is irradiated with ultraviolet rays **U** through the optical element **S4**.

(150) As illustrated in FIG. 5(d), an optical element laminate **2b** is obtained by curing the pressure-sensitive adhesive layer **10**. In FIG. 5(d), **10c** is a pressure-sensitive adhesive layer in which the pressure-sensitive adhesive layer **10** is cured.

(151) The crosslinking agent **11b** is crosslinked and polymerized to form a crosslinked structure **11c** by curing the pressure-sensitive adhesive layer **10**, thereby forming a pressure-sensitive adhesive layer **10c**. The pressure-sensitive adhesive layer **10c** has an improved elastic modulus and improved adhesive reliability to the substrate **20**. Therefore, the pressure-sensitive adhesive layer **10c** suppresses the generation of gas such as carbon dioxide due to heating of the substrate **20** (plastic film) and prevents the formation of bubbles.

(152) In the embodiment of FIG. 5(d), the crosslinking density on a side of the principal surface **10B**, in which the pressure-sensitive adhesive layer **10c** is in contact with the optical element **S4**, is higher than the principal surface **10A** on the opposite side. This configuration is preferable in that its bendability can be improved, for example, when the optical element laminate **2b** is used as a flexible image display device bending the side of the principal surface **10B** to the outside.

(153) In other words, when a flexible display is bent, tensile stress is generally applied to the outside while compressive stress is applied to the inside, with the outside stress being greater than the inside stress. Therefore, durability against bending can be improved by disposing the principal surface **10B** of the pressure-sensitive adhesive layer **10c** on the outside of the flexible display when it is bent.

(154) The optical element laminate **2b** in the present embodiment has, for example, the following configuration.

(155) The optical element laminate **2b** includes a substrate **20** comprising an optical element and a pressure-sensitive adhesive layer **10c**, wherein: the pressure-sensitive adhesive layer **10c** is laminated on a principal surface of a substrate **20** comprising the optical element; the pressure-sensitive adhesive layer **10c** is a single layer comprising a transparent base pressure-sensitive adhesive material cured by a reaction between the second polymerization initiator **11a** and the second crosslinking agent **11b**, and having two opposite principal surfaces; and in a case where the single pressure-sensitive adhesive layer is divided into two equal portions in a thickness direction, the density of a crosslinking structure **11c** in an area to which one of the two principal surfaces, a first principal surface **10A**, belongs is different from the density of the crosslinking structure **11c** in an area to which the other principal surface, a second principal surface **10B**, belongs.

(156) In the present embodiment, the pressure-sensitive adhesive layer **10c** is laminated so as to fill up a stepped space between the principal surface **20A** of the substrate **20** and the printed layer **30**.

EXAMPLES

(157) The present invention will be described in more detail below based on Examples, but the present invention is not limited to these Examples.

(158) (Preparation of Pressure-Sensitive Adhesive Sheet A)

(159) To a monomer mixture composed of 57 parts by weight of butyl acrylate (BA), 23 parts by weight of 4-hydroxybutyl acrylate (4HBA), 8 parts by weight of 2-hydroxyethyl acrylate (HEA), and 12 parts by weight of cyclohexyl acrylate (CHA) was incorporated 0.035 parts by weight of a photopolymerization initiator (trade name "Omnirad 184", manufactured by IGM Resins B.V.) and 0.035 parts by weight of a photopolymerization initiator (trade name "Omnirad 651", manufactured by IGM Resins B.V.). The resultant was then irradiated with ultraviolet rays until the viscosity thereof (measuring conditions: a BH viscometer No. 5 rotor; 10 rpm; and measuring temperature: 30° C.) turned to about 20 Pa·s to yield a prepolymer composition in which the monomer components were partially polymerized.

(160) Next, to the prepolymer composition was added 0.03 parts by weight of dipentaerythritol hexaacrylate (DPHA) and mixed to yield an acrylic pressure-sensitive adhesive composition. The above acrylic pressure-sensitive adhesive composition was applied onto a release-treated surface of a release film (trade name: "MRF #38", manufactured by Mitsubishi Plastics, Inc.) such that the thickness thereof after being formation of a pressure-sensitive adhesive layer becomes 250 μm to form a pressure-sensitive adhesive composition layer. Then, a release film (trade name: "MRN #38", manufactured by Mitsubishi Plastics, Inc.) was laminated onto a surface of the pressure-sensitive adhesive composition layer. Thereafter, the pressure-sensitive adhesive composition layer was light-cured by performing irradiation with ultraviolet rays under conditions of an illuminance of 5 mW/cm.² and a light quantity of 1500 mJ/cm.² to form a pressure-sensitive adhesive sheet A.

Example 1

(161) A release film was removed from one of opposite principal surfaces (referred to as "first surface") of the pressure-sensitive adhesive sheet A, and the exposed first surface was coated with an ethyl acetate solution prepared by diluting trimethylolpropane triacrylate (TMPTA) to a concentration of 80% by weight, a photopolymerization initiator (Omnirad 369E, manufactured by IGM Resins B.V.) to a concentration of 0.5% by weight by using a Wire Wound Rod type, #12 bar coater, manufactured by RD Specialties (target wet applying thickness: 27 μm). After application, the pressure-sensitive adhesive sheet A was heated and dried in an oven at 110° C. for 2 minutes, and then the solvent was volatilized and eliminated to yield a pressure-sensitive adhesive sheet A containing the pressure-sensitive adhesive layer in which TMPTA was dissolved.

(162) The chemical structural formula of a photopolymerization initiator (Omnirad 369E, 2-benzyl-2-dimethylamino-1-(4-morpholinophenyl)-butanone-1) was drawn using ChemDraw 19.0, and the calculated pKa determined using a computational module Chemoinformatics platform MOSES (Molecular Networks GmbH, Erlangen, Germany) was 7.68.

Example 2

(163) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator (Omnirad 369E, manufactured by IGM Resins B.V.) in the ethyl acetate solution was changed to 1.0% by weight.

Example 3

(164) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator (Omnirad 369E, manufactured by IGM Resins B.V.) in the ethyl acetate solution was changed to 2.0% by weight.

Example 4

(165) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 0.025% by weight using a photopolymerization initiator (Omnirad 907, manufactured by IGM Resins B.V.).

(166) The chemical structural formula of the photopolymerization initiator (Omnirad 907, 2-methyl-1-[4-(methylthio)phenyl]-2-morpholinopropan-1-one) was drawn using ChemDraw 19.0, and the calculated pKa determined using a computational module Chemoinformatics platform MOSES (Molecular Networks GmbH, Erlangen, Germany) was 5.56.

Example 5

(167) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 1.0% by weight using a photopolymerization initiator (Omnirad 907, manufactured by IGM Resins

B.V.).

Example 6

(168) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 2.0% by weight using a photopolymerization initiator (Omnirad 907, manufactured by IGM Resins B.V.).

Example 7

(169) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 4.0% by weight using a photopolymerization initiator (Omnirad 907, manufactured by IGM Resins B.V.).

Example 8

(170) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 8.0% by weight using a photopolymerization initiator (Omnirad 907, manufactured by IGM Resins B.V.).

Example 9

(171) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator (Omnirad 369E, manufactured by IGM Resins B.V.) in a toluene solution was changed to 2.0% by weight using toluene as a solvent.

Example 10

(172) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that a photopolymerization initiator (Omnirad 907, manufactured by IGM Resins B. V.) was used, and an ethyl acetate solution prepared by diluting the photopolymerization initiator to a concentration of 1.0% by weight was applied by using a Wire Wound Rod type, #3 bar coater, manufactured by RD Specialties (target wet applying thickness: 7 μm).

Example 11

(173) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that a photopolymerization initiator (Omnirad 907, manufactured by IGM Resins B. V.) was used, and an ethyl acetate solution prepared by diluting the photopolymerization initiator to a concentration of 1.0% by weight was applied by using a Wire Wound Rod type, #20 bar coater, manufactured by RD Specialties (target wet applying thickness: 46 μm).

Example 12

(174) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which trimethylolpropane triacrylate (TMPTA) was dissolved was obtained in the same manner as in Example 1, except that an ethyl acetate solution prepared by diluting TMPTA to a concentration of 80% by weight, a photopolymerization initiator (Omnirad 184, manufactured by IGM Resins B.V.) to a concentration of 2.0% by weight, and a hindered amine light stabilizer (Tinuvin 770DF, manufactured by BASF) to a concentration of 0.2% by weight was applied thereto.

(175) The chemical structural formula of the photopolymerization initiator (Omnirad 184, 1-hydroxycyclohexyl phenyl ketone) was drawn using ChemDraw 19.0, and the calculated pKa determined using a computational module Chemoinformatics platform MOSES (Molecular Networks GmbH, Erlangen, Germany) was 12.8.

(176) The chemical structural formula of the hindered amine light stabilizer (Tinuvin 770DF,

bis(2,2,6,6-tetramethyl-4-piperidyl) sebacate) was drawn using ChemDraw 19.0, and the calculated pKa determined using a computational module Chemoinformatics platform MOSES (Molecular Networks GmbH, Erlangen, Germany) was 9.47.

Example 13

(177) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 12, except that the concentration of the hindered amine light stabilizer in the ethyl acetate solution was changed to 0.5% by weight using the hindered amine light stabilizer (Tinuvin 770DF, manufactured by BASF).

Example 14

(178) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 12, except that the concentration of the hindered amine light stabilizer in the ethyl acetate solution was changed to 2.0% by weight using a hindered amine light stabilizer (Tinuvin 770DF, manufactured by BASF).

Comparative Example 1

(179) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 1.0% by weight using a photopolymerization initiator (Omnirad 184, manufactured by IGM Resins B.V.).

Comparative Example 2

(180) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 2.0% by weight using a photopolymerization initiator (Omnirad 184, manufactured by IGM Resins B.V.).

Comparative Example 3

(181) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 4.0% by weight using a photopolymerization initiator (Omnirad 184, manufactured by IGM Resins B.V.).

Comparative Example 4

(182) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 1.0% by weight using a photopolymerization initiator (Omnirad 127D, manufactured by IGM Resins B.V.).

(183) The chemical structural formula of the photopolymerization initiator (Omnirad 127D, 2-hydroxy-1-(4-(4-(2-hydroxy-2-methylpropionyl) benzyl)phenyl)-2-methylpropan-1-one) was drawn using ChemDraw 19.0, and the calculated pKa determined using a computational module Chemoinformatics platform MOSES (Molecular Networks GmbH, Erlangen, Germany) was 12.84.

Comparative Example 5

(184) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 2.0% by weight using a photopolymerization initiator (Omnirad 127D, manufactured by IGM Resins B.V.).

Comparative Example 6

(185) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 1, except that the

concentration of the photopolymerization initiator in the ethyl acetate solution was changed to 4.0% by weight using a photopolymerization initiator (Omnirad 127D, manufactured by IGM Resins B.V.).

Comparative Example 7

(186) A pressure-sensitive adhesive sheet A containing a pressure-sensitive adhesive layer in which TMPTA was dissolved was obtained in the same manner as in Example 12, except that the concentration of the hindered amine light stabilizer in the ethyl acetate solution was changed to 0.5% by weight using a hindered amine light stabilizer (Tinuvin 123, manufactured by BASF).

(187) The chemical structural formula of the hindered amine light stabilizer (Tinuvin 123, bis(2,2,6,6-tetramethyl-1(octyloxy)-4-piperidinyl) ester) was drawn using ChemDraw 19.0, and the calculated pKa determined using a computational module Chemoinformatics platform MOSES (Molecular Networks GmbH, Erlangen, Germany) was 4.94.

(188) <Evaluation of Residual Stress Change Rate>

(189) The release films of the pressure-sensitive adhesive sheets obtained in Examples 1 to 14 and Comparative Examples 1 to 7 were each peeled off to prepare test pieces having a width of 40 mm and a length of 20 mm. The test pieces were rolled back in the longitudinal direction to prepare cylindrical test pieces (the length of each cylinder is 40 mm). At this time, the test pieces were prepared so as not to have voids inside. Both ends of each cylinder in the longitudinal direction were held with chucks of an autograph (AG-X plus 200N, manufactured by Shimadzu Corporation) by 10 mm (the distance between the chucks is 20 mm). The test pieces were each pulled by the autograph until the distance between the chucks reached 80 mm, and held for 300 seconds while the distance between the chucks remained at 80 mm. The residual stress [N/cm.sup.2] was obtained by the formula [stress after holding for 300 seconds/length×glue thickness].

(190) The residual stress at the initial stage of production of the pressure-sensitive adhesive sheet and the residual stress of the pressure-sensitive adhesive sheet after storage in a thermostatic bath at 50° C. for 4 weeks were determined, and the rate (%) of change in residual stress was calculated from the following equation.

Residual stress change rate (%)=(residual stress after storage at 50° C. for 4 weeks–initial residual stress)/(initial residual stress)×100

(191) The pressure-sensitive adhesive sheets were evaluated for the storage stability based on the following criteria. Results are shown in Table 1. ⊙: Residual stress change rate of 30% or less ○: Residual stress change rate of more than 30% and not more than 50% Δ: Residual stress change rate of more than 50% and not more than 70% X: Residual stress change rate of more than 70% and not more than 100% XX: Residual stress change rate of more than 100%

(192) TABLE-US-00001 TABLE 1 Residual stress change rate (%) stability Example rate (%) stability 1 26.5 ⊙ 1 117.2 XX 2 17.3 ⊙ 2 106.6 XX 3 17.3 ⊙ 3 103.9 XX 4 28.3 ⊙ 4 98.1 X 5 44.6 ○ 5 82.5 X 6 37.0 ○ 6 74.3 X 7 24.7 ⊙ 7 124.6 XX 8 21.2 ⊙ 9 17.3 ⊙ 10 45.9 ○ 11 42.4 ○ 12 24.6 ⊙ 13 58.3 Δ 14 49.7 ○

(193) Variations of the present invention will be supplementally described below.

(194) [Supplement 1]

(195) A radiation curable pressure-sensitive adhesive sheet having a pressure-sensitive adhesive layer that is cured by irradiation with radiation, wherein: the pressure-sensitive adhesive layer contains a photopolymerization initiator and a crosslinking agent; the curing by irradiation with radiation is curing by a reaction between the photopolymerization initiator and the crosslinking agent; and when the pressure-sensitive adhesive layer is stored at 50° C. for 4 weeks, a rate of change in residual stress (N/cm.sup.2) expressed by the following equation is 70% or less: residual stress change rate (%)=(residual stress after storage at 50° C. for 4 weeks–initial residual stress)/(initial residual stress)×100.

[Supplement 2]

(196) The radiation curable pressure-sensitive adhesive sheet according to supplement 1, having a

pressure-sensitive adhesive layer that is cured by irradiation with radiation, wherein: the pressure-sensitive adhesive layer is cured by a reaction between a first photopolymerization initiator and a first crosslinking agent; the pressure-sensitive adhesive layer contains a second photopolymerization initiator and a second crosslinking agent; the curing by irradiation with radiation is curing by a reaction between the second photopolymerization initiator and the second crosslinking agent; and the second photopolymerization initiator has a pKa of 5 to 12.7.

[Supplement 3]

(197) The radiation curable pressure-sensitive adhesive sheet according to supplement 2, wherein the second photopolymerization initiator and the second crosslinking agent are dissolved in the pressure-sensitive adhesive layer.

(198) [Supplement 4]

(199) The radiation curable pressure-sensitive adhesive sheet according to supplement 2 or 3, wherein the second photopolymerization initiator is the same as the first photopolymerization initiator.

(200) [Supplement 5]

(201) The radiation curable pressure-sensitive adhesive sheet according to any one of supplements 1 to 4, wherein: the pressure-sensitive adhesive layer is a single layer having two opposite principal surfaces; in a case where the single pressure-sensitive adhesive layer is divided into two equal portions in a thickness direction, a concentration of the photopolymerization initiator (second photopolymerization initiator) and the crosslinking agent (second crosslinking agent) in an area to which one of the two principal surfaces, a first principal surface, belongs is different from a concentration of the photopolymerization initiator (second photopolymerization initiator) and the crosslinking agent (second crosslinking agent) in an area to which the other principal surface, a second principal surface, belongs.

[Supplement 6]

(202) The radiation curable pressure-sensitive adhesive sheet according to supplement 5, wherein the single pressure-sensitive adhesive layer has a concentration gradient of the photopolymerization initiator (second photopolymerization initiator) and the crosslinking agent (second crosslinking agent) in the thickness direction.

(203) [Supplement 7]

(204) The radiation curable pressure-sensitive adhesive sheet according to any one of supplements 1 to 6, wherein the pressure-sensitive adhesive layer has a thickness of 5 to 500 μm .

(205) [Supplement 8]

(206) A method for producing the radiation curable pressure-sensitive adhesive sheet according to any one of supplements 1 to 7, comprising: forming a pressure-sensitive adhesive layer formed of a base pressure-sensitive adhesive material on a support; curing the pressure-sensitive adhesive layer; providing a solution of a photopolymerization initiator (second photopolymerization initiator) and a crosslinking agent (second crosslinking agent); applying the solution to one of opposite surfaces of the cured pressure-sensitive adhesive layer; causing the photopolymerization initiator (second photopolymerization initiator) and the crosslinking agent (second crosslinking agent) contained in the solution to infiltrate from the one surface in a thickness direction of the pressure-sensitive adhesive layer; and drying the pressure-sensitive adhesive layer.

[Supplement 9]

(207) The method for producing the radiation curable pressure-sensitive adhesive sheet according to supplement 8, wherein the solution containing the photopolymerization initiator (second photopolymerization initiator) and the crosslinking agent (second crosslinking agent) is a solution in which the photopolymerization initiator and the crosslinking agent are dissolved in a solvent, and the method comprises drying the pressure-sensitive adhesive layer to evaporate the solvent of the solution.

[Supplement 10]

(208) The method for producing the radiation curable pressure-sensitive adhesive sheet according to supplement 8 or 9, wherein the support is a release sheet.

(209) [Supplement 11]

(210) The method for producing a radiation curable pressure-sensitive adhesive sheet according to any one of supplements 8 to 10 further comprising laminating a release sheet onto a surface of the pressure-sensitive adhesive layer on a side opposite to the support.

(211) [Supplement 12]

(212) An optical element laminate comprising: a substrate comprising an optical element; and a pressure-sensitive adhesive layer,

wherein: the pressure-sensitive adhesive layer is laminated on a principal surface of the substrate comprising an optical element; and the pressure-sensitive adhesive layer is a cured product of the pressure-sensitive adhesive layer of the radiation curable pressure-sensitive adhesive sheet according to any one of supplements 1 to 7.

[Supplement 13]

(213) The optical element laminate according to supplement 12, wherein: the principal surface of the substrate comprising an optical element has a printed layer; and the pressure-sensitive adhesive layer is laminated so as to fill up a stepped space between the principal surface of the substrate comprising an optical element and the printed layer.

[Supplement 14]

(214) The optical element laminate according to supplement 12 or 13, wherein the cured product is a cured product obtained by reaction of the polymerization initiator and a crosslinking agent.

INDUSTRIAL APPLICABILITY

(215) A radiation curable pressure-sensitive adhesive sheet of the present invention is useful in a pressure-sensitive adhesive sheet having a transparent pressure-sensitive adhesive layer that can be used for bonding a transparent optical element to another optical element.

REFERENCE SIGNS LIST

(216) **1** Radiation curable pressure-sensitive adhesive sheet **1a** Radiation curable pressure-sensitive adhesive sheet **1b** Radiation curable pressure-sensitive adhesive sheet **2** Optical element laminate **2a** Optical element laminate **2b** Optical element laminate **10a** Pressure-sensitive adhesive layer (before curing) **10** Pressure-sensitive adhesive layer (after curing) **10c** Pressure-sensitive adhesive layer (after curing) **S1, S2** Support (release sheet) **U** Ultraviolet ray **11a** Photopolymerization initiator **11b** Crosslinking agent **11c** Crosslinked structure **12** Solution of photopolymerization initiator and crosslinking agent **13** Solvent **S3, S4** Optical element **20** Substrate (optical element) **30** Printed layer

Claims

1. A pressure-sensitive adhesive sheet, comprising: i) a pressure-sensitive adhesive layer, the pressure-sensitive adhesive layer comprising, in polymerized form, a base monomer, a first photopolymerization initiator and a first crosslinking agent, the pressure-sensitive adhesive layer having two opposite principal surfaces: first principal surface and second principal surface; ii) a second photopolymerization initiator; iii) a second crosslinking agent, and iv) a release sheet laminated onto the first principal surface of the pressure-sensitive adhesive layer, wherein: when the pressure-sensitive adhesive layer is stored at 50° C. for 4 weeks, a rate of change in residual stress (N/cm.sup.2) expressed by the following equation is 70% or less:
$$\text{residual stress change rate (\%)} = \frac{(\text{residual stress after storage at 50}^\circ \text{C. for 4 weeks} - \text{initial residual stress})}{(\text{initial residual stress})} \times 100$$
, and the first crosslinking agent forms a crosslinked structure in the pressure-sensitive adhesive layer, while the second crosslinking agent is an unreacted crosslinking agent and is present within the pressure-sensitive adhesive layer and the second photopolymerization initiator is an unreacted photopolymerization initiator and is present within

the pressure-sensitive adhesive layer.

2. The pressure-sensitive adhesive sheet according to claim 1, wherein the second photopolymerization initiator has a pKa of 5 to 12.7.

3. The pressure-sensitive adhesive sheet according to claim 1, wherein the second photopolymerization initiator and the second crosslinking agent are dissolved in the pressure-sensitive adhesive layer.

4. The pressure-sensitive adhesive sheet according to claim 1, wherein the second photopolymerization initiator and the first photopolymerization initiator are the same type of a photopolymerization initiator.

5. The pressure-sensitive adhesive sheet according to claim 1, wherein: the pressure-sensitive adhesive layer is a single layer having the two opposite principal surfaces; in a case where the single pressure-sensitive adhesive layer is divided into two equal portions in a thickness direction, a concentration of the second photopolymerization initiator and the second crosslinking agent in an area to which one of the two opposite principal surfaces, the first principal surface, belongs is different from a concentration of the second photopolymerization initiator and the second crosslinking agent in an area to which the other principal surface, the second principal surface, belongs.

6. The pressure-sensitive adhesive sheet according to claim 5, wherein the single pressure-sensitive adhesive layer has a concentration gradient of the second photopolymerization initiator and the second crosslinking agent in the thickness direction.

7. The pressure-sensitive adhesive sheet according to claim 1, wherein the pressure-sensitive adhesive layer has a thickness of 5 to 500 μm .

8. A method for producing the pressure-sensitive adhesive sheet according to claim 1, the method comprising: forming a pressure-sensitive adhesive layer formed of a base pressure-sensitive adhesive material on a support; curing the pressure-sensitive adhesive layer; providing a solution of a second photopolymerization initiator and a second crosslinking agent; applying the solution to one of opposite surfaces of the cured pressure-sensitive adhesive layer; causing the second photopolymerization initiator and the second crosslinking agent contained in the solution to infiltrate from the one surface in a thickness direction of the pressure-sensitive adhesive layer; and drying the pressure-sensitive adhesive layer.

9. The method for producing the pressure-sensitive adhesive sheet according to claim 8, wherein the solution containing the second photopolymerization initiator and the second crosslinking agent is a solution in which the second photopolymerization initiator and the second crosslinking agent are dissolved in a solvent, and the method comprises drying the pressure-sensitive adhesive layer to evaporate the solvent of the solution.

10. The method for producing the pressure-sensitive adhesive sheet according to claim 8, wherein the support is a release sheet.

11. The method for producing a pressure-sensitive adhesive sheet according to claim 8, further comprising laminating a release sheet onto a surface of the pressure-sensitive adhesive layer on a side opposite to the support.

12. An optical element laminate comprising: a substrate comprising an optical element; and a pressure-sensitive adhesive layer, wherein: the pressure-sensitive adhesive layer is laminated on a principal surface of the substrate comprising an optical element; and the pressure-sensitive adhesive layer is a cured product of the pressure-sensitive adhesive layer of the pressure-sensitive adhesive sheet according to claim 1.

13. The optical element laminate according to claim 12, wherein: the principal surface of the substrate comprising an optical element has a printed layer; and the pressure-sensitive adhesive layer is laminated so as to fill up a stepped space between the principal surface of the substrate comprising an optical element and the printed layer.

14. The optical element laminate according to claim 12, wherein the cured product is a cured product obtained by reaction of the polymerization initiator and a crosslinking agent.
